



US012396429B2

(12) **United States Patent**
Wiebbecke

(10) **Patent No.:** **US 12,396,429 B2**

(45) **Date of Patent:** ***Aug. 26, 2025**

(54) **SOYBEAN VARIETY 01098164**

(71) Applicant: **Monsanto Technology LLC**, St. Louis,
MO (US)

(72) Inventor: **Christiana E. Wiebbecke**, Baxter, IA
(US)

(73) Assignee: **Monsanto Technology LLC**, St. Louis,
MO (US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 208 days.

This patent is subject to a terminal dis-
claimer.

(21) Appl. No.: **17/992,417**

(22) Filed: **Nov. 22, 2022**

(65) **Prior Publication Data**

US 2024/0164327 A1 May 23, 2024

(51) **Int. Cl.**

A01H 6/54 (2018.01)

A01H 5/10 (2018.01)

(52) **U.S. Cl.**

CPC **A01H 6/542** (2018.05); **A01H 5/10**
(2013.01)

(58) **Field of Classification Search**

None

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,850,019 A 12/1998 Maiti et al.
6,051,753 A 4/2000 Comai et al.
6,660,911 B2 12/2003 Fincher et al.
6,949,696 B2 9/2005 Fincher et al.
7,141,722 B2 11/2006 Fincher et al.
7,608,761 B2 10/2009 Baley et al.
7,632,985 B2 12/2009 Malven et al.
7,812,224 B2 10/2010 Weeks et al.
7,838,729 B2 11/2010 Feng et al.
7,855,326 B2 12/2010 Feng et al.
7,884,262 B2 2/2011 Clemente et al.
7,939,721 B2 5/2011 Arnevik et al.
8,017,756 B2 9/2011 De Beuckeleer
8,053,184 B2 11/2011 Malven et al.
8,119,380 B2 2/2012 Weeks et al.
8,207,092 B2 6/2012 Bhatti et al.
8,501,407 B2 8/2013 Brinker et al.
8,629,323 B2 1/2014 Weeks et al.
8,700,336 B2 4/2014 Van Den Bulcke et al.
8,754,011 B2 6/2014 Bhatti et al.
RE45,048 E 7/2014 Feng et al.
8,952,142 B2 2/2015 De Beuckeleer
9,017,947 B2 4/2015 Malven et al.

OTHER PUBLICATIONS

Allard, In: Principles of Plant Breeding, Chapter 6 through Chapter
9, University of California, Davis, California, John Wiley & Sons,
New York, pp. 50-98, 1960.

Eshed et al., "Less-Than-Addictive Epistatic Interactions of Quan-
titative Trait Loci in Tomato," Genetics, 143:1807-1817, 1996.

Fehr, "Breeding Methods for Cultivar Development," In: Soybeans:
Improvement, Production and Uses, 2nd Edition, Wilcox et al.,
(Eds.), Madison, Wisconsin, 16, pp. 249 and 259, 1987.

Fehr (ed), "Backcross Method," In: Principles of Cultivar Devel-
opment, vol. 1: Theory and Technique, pp. 360-376, 1987.

Kraft et al., Linkage Disequilibrium and Fingerprinting in Sugar
Beet, Theor Appl Genet, 101:323-326, 2000.

Variety specific information as indicated in transmittal letter of the
Information Disclosure Statement for U.S. Appl. No. 17/992,417,
dated Jan. 21, 2025.

2018, The cross was made near Juana Diaz, PR, USA.

2019, Plants were grown near Juana Diaz, PR, USA. A backcross
was made using an F1 single plant selection and BN2919C1-
CODNN as the recurrent parent.

2019, Plants were grown near Juana Diaz, PR, USA. A backcross
was made using an BC1F1 single plant selection and BN2919C1-
CODNN as the recurrent parent.

2019, Plants were grown near Juana Diaz, PR, USA. A backcross
was made using an BC2F1 single plant selection and BN2919C1-
CODNN as the recurrent parent.

2019, Plants were grown near Juana Diaz, PR, USA and advanced
using single plant selection.

2020, Plants were grown near Juana Diaz, PR, USA in Progeny
Rows and the variety 01098164 was selected based on the agro-
nomic characteristics, general phenotypic appearance and traits of
interest based on molecular marker information.

Advancement/Selection Criteria.

2021, Yield, Agronomics, Disease, and Quality.

* cited by examiner

Primary Examiner — Brent T Page

(74) *Attorney, Agent, or Firm* — Paul D. Tietz; Ann H.
Inglett

(57) **ABSTRACT**

The invention relates to the soybean variety designated
01098164. Provided by the invention are the seeds, plants
and derivatives of the soybean variety 01098164. Also
provided by the invention are tissue cultures of the soybean
variety 01098164 and the plants regenerated therefrom. Still
further provided by the invention are methods for producing
soybean plants by crossing the soybean variety 01098164
with itself or another soybean variety and plants produced
by such methods.

20 Claims, No Drawings

1

SOYBEAN VARIETY 01098164

BACKGROUND OF THE INVENTION

Field of the Invention

The present invention relates generally to the field of soybean breeding. In particular, the invention relates to the novel soybean variety 01098164, and more particularly to soybean plants and seeds of this new variety.

Description of Related Art

There are numerous steps in the development of any novel plant germplasm. Plant breeding begins with the analysis and definition of problems and weaknesses of the current germplasm, the establishment of program goals, and the definition of specific breeding objectives. The next step is selection of germplasm that possess the traits to meet the program goals. The goal is to combine in a single variety an improved combination of traits from the parental germplasm. These important traits may include higher seed yield, resistance to diseases and insects, better stems and roots, tolerance to drought and heat, better agronomic quality, resistance to herbicides, and improvements in compositional traits.

Soybean, *Glycine max* (L.), is a valuable field crop. Thus, a goal of plant breeders is to develop stable, high-yielding soybean varieties that are agronomically sound. The reasons for this goal are to maximize the amount of grain produced on the land used and to supply food for both animals and humans. To accomplish this goal, the soybean breeder must select and develop soybean plants that have the traits that result in superior varieties.

The oil extracted from soybeans is widely used in food products, such as margarine, cooking oil, and salad dressings. Soybean oil is composed of saturated, monounsaturated, and polyunsaturated fatty acids, with a typical composition of 11% palmitic, 4% stearic, 25% oleic, 50% linoleic, and 9% linolenic fatty acid content ("Economic implications of modified soybean traits: summary report," Iowa Soybean Promotion Board, Soybean Trait Modification Task Force, and American Soybean Association Special Report 92S, May 1990).

SUMMARY OF THE INVENTION

One aspect of the present invention relates to seed of the soybean variety 01098164. The invention also relates to plants produced by growing the seed of the soybean variety 01098164, as well as the derivatives of such plants. Further provided are plant parts, including cells, plant protoplasts, plant cells of a tissue culture from which soybean plants can be regenerated, plant calli, plant clumps, and plant cells that are intact in plants or plant parts, such as pollen, flowers, seeds, pods, leaves, stems, and the like.

In a further aspect, the invention provides a composition comprising a seed of soybean variety 01098164 comprised in plant seed growth media. In certain embodiments, the plant seed growth media is a soil or synthetic cultivation medium. In specific embodiments, the growth medium may be comprised in a container or may, for example, be soil in a field. Plant seed growth media are well known to those of skill in the art and include, but are in no way limited to, soil or synthetic cultivation medium. Plant seed growth media can provide adequate physical support for seeds and can retain moisture and/or nutritional components. Examples of

2

characteristics for soils that may be desirable in certain embodiments can be found, for instance, in U.S. Pat. Nos. 3,932,166 and 4,707,176. Synthetic plant cultivation media are also well known in the art and may, in certain embodiments, comprise polymers or hydrogels. Examples of such compositions are described, for example, in U.S. Pat. No. 4,241,537.

Another aspect of the invention relates to a tissue culture of regenerable cells of the soybean variety 01098164, as well as plants regenerated therefrom, wherein the regenerated soybean plant is capable of expressing all the morphological and physiological characteristics of a plant grown from the soybean seed designated 01098164.

Yet another aspect of the current invention is a soybean plant further comprising a single locus conversion. In one embodiment, the soybean plant is defined as further comprising the single locus conversion and otherwise capable of expressing all of the morphological and physiological characteristics of the soybean variety 01098164. In particular embodiments of the invention, the single locus conversion may comprise a transgenic gene which has been introduced by genetic transformation into the soybean variety 01098164 or a progenitor thereof. In still other embodiments of the invention, the single locus conversion may comprise a dominant or recessive allele. The locus conversion may confer potentially any trait upon the single locus converted plant, including herbicide resistance, insect resistance, resistance to bacterial, fungal, or viral disease, male fertility or sterility, and improved nutritional quality. In certain embodiments, a potential locus conversion that confers herbicide resistance may confer resistance to herbicides such as, for example, imidazolinone herbicides, sulfonylurea herbicides, triazine herbicides, phenoxy herbicides, cyclohexanedione herbicides, benzonitrile herbicides, 4-hydroxyphenylpyruvate dioxygenase-inhibiting herbicides, protoporphyrinogen oxidase-inhibiting herbicides, acetolactate synthase-inhibiting herbicides, 1-aminocyclopropane-1-carboxylic acid synthase-inhibiting herbicides, bromoxynil, nicosulfuron, 2,4-dichlorophenoxyacetic acid (2,4-D), dicamba, quizalofop-p-ethyl, glyphosate, or glufosinate.

In further embodiments, a single locus conversion comprises a genetic modification to the genome of soybean variety 01098164. A genetic modification may comprise, for example, an insertion, deletion, or substitution of a nucleotide sequence. In certain embodiments, a single locus may comprise one or more genes or intergenic regions integrated into or mutated at a single locus or may comprise one or more nucleic acid molecules integrated at the single locus. In particular embodiments, a single locus conversion may be generated by genome editing such as through use of engineered nucleases, as is known in the art. Examples of engineered nucleases include, but are not limited to, Cas endonucleases, zinc finger nucleases (ZFNs), transcription activator-like effector nucleases (TALENs), and engineered meganucleases, also known as homing endonucleases. Naturally occurring nucleases can also find use for genome editing. In specific embodiments, endonucleases, both naturally occurring and engineered, may utilize any polypeptide-, DNA-, or RNA-guided genome editing systems known to the skilled artisan.

Still yet another aspect of the invention relates to a first generation (F_1) hybrid soybean seed produced by crossing a plant of the soybean variety 01098164 to a second soybean plant. Also included in the invention are the F_1 hybrid soybean plants grown from the hybrid seed produced by crossing the soybean variety 01098164 to a second soybean plant.

3

Still yet another aspect of the invention is a method of producing soybean seeds comprising crossing a plant of the soybean variety 01098164 to any second soybean plant, including itself or another plant of the variety 01098164. In particular embodiments of the invention, the method of crossing comprises the steps of a) planting seeds of the soybean variety 01098164; b) cultivating soybean plants resulting from said seeds until said plants bear flowers; c) allowing fertilization of the flowers of said plants; and d) harvesting seeds produced from said plants.

Still yet another aspect of the invention is a method of producing hybrid soybean seeds comprising crossing the soybean variety 01098164 to a second, distinct soybean plant that is nonisogenic to the soybean variety 01098164. In particular embodiments of the invention, the crossing comprises the steps of a) planting seeds of soybean variety 01098164 and a second, distinct soybean plant, b) cultivating the soybean plants grown from the seeds until the plants bear flowers; c) cross-pollinating a flower on one of the two plants with the pollen of the other plant, and d) harvesting the seeds resulting from the cross-pollinating.

Still yet another aspect of the invention is a method for developing a soybean plant in a soybean breeding program comprising: obtaining a soybean plant, or its parts, of the variety 01098164; and b) employing said plant or parts as a source of breeding material using plant breeding techniques. In the method, the plant breeding techniques may be selected from the group consisting of recurrent selection, mass selection, bulk selection, backcrossing, pedigree breeding, genetic marker-assisted selection and genetic transformation. In certain embodiments of the invention, the soybean plant of variety 01098164 is used as the male or female parent.

Still yet another aspect of the invention is a method of producing a soybean plant derived from the soybean variety

4

second plant to produce a progeny plant of a further subsequent generation that is derived from a plant of the soybean variety 01098164; and (d) repeating step (c), in some embodiments, at least 1, 2, 3, 4 or more additional generations to produce an inbred soybean plant that is derived from the soybean variety 01098164. The invention still further provides a soybean plant produced by this and the foregoing methods.

In another embodiment of the invention, the method of producing a soybean plant derived from the soybean variety 01098164 further comprises: (a) crossing the soybean variety 01098164-derived soybean plant with itself or another soybean plant to yield additional soybean variety 01098164-derived progeny soybean seed; (b) growing the progeny soybean seed of step (a) under plant growth conditions to yield additional soybean variety 01098164-derived soybean plants; and (c) repeating the crossing and growing steps of (a) and (b) to generate further soybean variety 01098164-derived soybean plants. In specific embodiments, steps (a) and (b) may be repeated at least 1, 2, 3, 4, or 5 or more times as desired. The invention still further provides a soybean plant produced by this and the foregoing methods.

DETAILED DESCRIPTION OF THE INVENTION

The instant invention provides methods and composition relating to plants, seeds and derivatives of the soybean variety 01098164. Soybean variety 01098164 is adapted to LATE GROUP II. Soybean variety 01098164 was developed from an initial cross of BN2919C1-C0DNN/WO4215B4-T3LNN. The breeding history of the variety can be summarized as follows:

Generation	Year	Description
Cross	2018	The cross was made near Juana Diaz, PR, USA
BC1	2019	Plants were grown near Juana Diaz, PR, USA. A backcross was made using an F1 single plant selection and BN2919C1-C0DNN as the recurrent parent.
BC2	2019	Plants were grown near Juana Diaz, PR, USA. A backcross was made using an BC1F1 single plant selection and BN2919C1-C0DNN as the recurrent parent.
BC3	2019	Plants were grown near Juana Diaz, PR, USA. A backcross was made using an BC2F1 single plant selection and BN2919C1-C0DNN as the recurrent parent.
BC3F1	2019	Plants were grown near Juana Diaz, PR, USA and advanced using single plant selection.
BC3F2	2020	Plants were grown near Juana Diaz, PR, USA in Progeny Rows and the variety 01098164 was selected based on the agronomic characteristics, general phenotypic appearance and traits of interest based on molecular marker information.
Yield Testing		
Generation	Year	Advancement/Selection Criteria
BC3F3	2021	Yield, Agronomics, Disease, and Quality

01098164, the method comprising the steps of: (a) crossing a plant of the soybean variety 01098164 with a second soybean plant to produce a progeny plant that is derived from soybean variety 01098164; and (b) crossing the progeny plant with itself or a second plant to produce a progeny plant of a subsequent generation that is derived from a plant of the soybean variety 01098164. In one embodiment of the invention, the method further comprises: (c) crossing the progeny plant of a subsequent generation with itself or a

The soybean variety 01098164 has been judged to be uniform for breeding purposes and testing. The variety 01098164 can be reproduced by planting and growing seeds of the variety under self-pollinating or sib-pollinating conditions, as is known to those of skill in the agricultural arts. Variety 01098164 shows no variants other than what would normally be expected due to environment or that would occur for almost any characteristic during the course of repeated sexual reproduction.

The results of an objective evaluation of the variety are presented below, in Table 1. Those of skill in the art will recognize that these are typical values that may vary due to environment and that other values that are substantially equivalent are within the scope of the invention. An “*” denotes classifications/scores generated based on greenhouse assays.

TABLE 1

Phenotypic Description of Variety 01098164	
Trait	VALUE
Morphology:	
Relative Maturity	2.8
Flower Color	PURPLE
Pubescence Color	GRAY
Hilum Color	IMPERFECT BLACK
Pod Color	BROWN
Seed Coat Color	YELLOW
Seed Coat Luster	DULL
Seed Shape	SPHERICAL FLATTENED
Cotyledon Color	YELLOW
Leaf Shape	OVATE
Leaf Color	GREEN
Canopy	INTERMEDIATE
Growth Habit	INDETERMINATE
Disease Reactions:	
Phytophthora Allele*	RPS1C
Phytophthora Tolerance*	MODERATELY TOLERANT
Soybean Cyst Nematode Race 3*	RESISTANT
Brown Stem Rot*	RESISTANT
Southern Stem Canker*	RESISTANT
Herbicide Reactions:	
Glyphosate	RESISTANT, MON89788
Glufosinate	RESISTANT A5547-127
Dicamba	RESISTANT, MON87708

As disclosed herein above, soybean variety 01098164 contains events MON89788, A5547-127, and MON87708. Event MON89788, also known as event GM_A19788, confers glyphosate tolerance and is the subject of U.S. Pat. No. 7,632,985, the disclosure of which is incorporated herein by reference. Event MON89788 is also covered by one or more of the following patents: U.S. Pat. Nos. 6,051,753; 6,660,911; 6,949,696; 7,141,722; 7,608,761; 8,053,184; and 9,017,947. Event A5547-127, also known as event EE-GM2, event LL55, and event ACS-GM006-4, confers glufosinate tolerance and is the subject of U.S. Pat. No. 8,017,756, the disclosure of which is incorporated herein by reference. Event A5547-127 is also covered by one or more of the following patents: U.S. Pat. Nos. 8,700,336; 8,952,142; 9,062,324; and 9,683,242. Event MON87708 confers dicamba tolerance and is the subject of U.S. Pat. No. 8,501,407, the disclosure of which is incorporated herein by reference. Event MON87708 is also covered by one or more of the following patents: U.S. Pat. Nos. 5,850,019; 7,812,224; 7,838,729; 7,884,262; 7,939,721; 8,119,380; 8,207,092; 8,629,323; 8,754,011; and RE45,048.

Breeding Soybean Variety 01098164

One aspect of the current invention concerns methods for crossing the soybean variety 01098164 with itself or a second plant and the seeds and plants produced by such methods. These methods can be used for propagation of the soybean variety 01098164, or can be used to produce hybrid soybean seeds and the plants grown therefrom. Hybrid soybean plants can be used by farmers in the commercial

production of soy products or may be advanced in certain breeding protocols for the production of novel soybean varieties. A hybrid plant can also be used as a recurrent parent at any given stage in a backcrossing protocol during the production of a single locus conversion of the soybean variety 01098164.

Soybean variety 01098164 is well suited to the development of new varieties based on the elite nature of the genetic background of the variety. In selecting a second plant to cross with 01098164 for the purpose of developing novel soybean varieties, it will typically be desired to choose those plants that either themselves exhibit one or more selected characteristics or that exhibit the characteristic(s) when in hybrid combination. Examples of potentially selected characteristics include seed yield, lodging resistance, emergence, seedling vigor, disease tolerance, maturity, plant height, high oil content, high protein content and shattering resistance.

Choice of breeding or selection methods depends on the mode of plant reproduction, the heritability of the trait(s) being improved, and the type of variety used commercially (e.g., F₁ hybrid variety, pureline variety, etc.). For highly heritable traits, a choice of superior individual plants evaluated at a single location will be effective; whereas, for traits with low heritability, selection should be based on mean values obtained from replicated evaluations of families of related plants. Popular selection methods commonly include pedigree selection, modified pedigree selection, mass selection, recurrent selection and backcrossing.

The complexity of inheritance influences the choice of the breeding method. Backcross breeding is used to transfer one or a few genes for a highly heritable trait into a desirable variety. This approach has been used extensively for breeding disease-resistant varieties (Bowers et al., *Crop Sci.*, 32(1):67-72, 1992; Nickell and Bernard, *Crop Sci.*, 32(3):835, 1992). Various recurrent selection techniques are used to improve quantitatively inherited traits controlled by numerous genes. The use of recurrent selection in self-pollinating crops depends on the ease of pollination, the frequency of successful hybrids from each pollination, and the number of hybrid offspring from each successful cross.

Each breeding program should include a periodic, objective evaluation of the efficiency of the breeding procedure. Evaluation criteria vary depending on the goal and objectives, but should include gain from selection per year based on comparisons to an appropriate standard, overall value of the advanced breeding lines, and number of successful varieties produced per unit of input, e.g., per year, per dollar expended, etc.

Promising advanced breeding lines are thoroughly tested and compared to appropriate standards in environments that are representative of the commercial target area(s) for generally three or more years. The best lines are candidates for new commercial varieties. Those still deficient in a few traits may be used as parents to produce new populations for further selection.

These processes, which lead to the final step of marketing and distribution, may take as much as eight to 12 years from the time the first cross is made. Therefore, development of new varieties is a time-consuming process that requires precise forward planning, efficient resource utilization, and minimal direction changes.

Identifying individuals that are genetically superior is a difficult task because the true genotypic value for most traits can be masked by other confounding traits or environmental factors. One method of identifying a superior plant is observing its performance relative to other experimental plants and one or more widely grown standard varieties.

Single observations are generally inconclusive, while replicated observations provide a better estimate of genetic worth.

The goal of plant breeding is to develop new, unique, and superior soybean varieties and hybrids. The breeder initially selects and crosses two or more parental lines. This is generally followed by repeated selfing and selection, which produces many new genetic combinations. Each year, the plant breeder selects the germplasm to advance to the next generation. This germplasm is grown under unique and different geographical, climatic, and soil conditions, and further selections are then made during and at the end of the growing season. The varieties which are developed are unpredictable. This unpredictability is because the breeder's selection occurs in unique environments, with no control at the DNA level (using conventional breeding procedures), and with millions of different possible genetic combinations being generated. A breeder of ordinary skill in the art cannot predict the final resulting lines he develops, except possibly in a gross and general fashion. The same breeder cannot produce the same variety twice by using the exact same original parents and the same selection techniques. This unpredictability results in the expenditure of large amounts of research monies to develop superior new soybean varieties.

Pedigree breeding and recurrent selection breeding methods are used to develop varieties from breeding populations. Breeding programs combine traits from two or more varieties or various broad-based sources into breeding pools from which varieties are developed by selfing and selection of phenotypes. The new varieties are evaluated to determine which have commercial potential.

Pedigree breeding is commonly used for the improvement of self-pollinating crops. Two parents which possess favorable, complementary traits are crossed to produce F_1 progeny. An F_2 population is then produced by selfing one or several F_1 plants. Selection of the best individuals may begin in the F_2 population or later depending upon the breeder's objectives; then, beginning in the F_3 generation, the best individuals in the best families can be selected. Replicated testing of families can begin in the F_3 or F_4 generations to improve the effectiveness of selection for traits of low heritability. At an advanced stage of inbreeding (i.e., the F_6 and F_7 generations), the best lines or mixtures of phenotypically similar lines are tested for potential release as new varieties.

Mass and recurrent selections can be used to improve populations of either self- or cross-pollinating crops. A genetically variable population of heterozygous individuals is either identified or created by intercrossing several different parents. The best plants are selected based on individual superiority, outstanding progeny, or excellent combining ability. The selected plants are intercrossed to produce a new population from which further cycles of selection are continued.

Backcross breeding has been used to transfer genetic loci for simply inherited or highly heritable traits into a homozygous variety that is used as the recurrent parent. The source of the trait to be transferred is called the donor or nonrecurrent parent. The resulting plant is expected to have the attributes of the recurrent parent and the trait transferred from the donor parent. After the initial cross, individuals possessing the phenotype of the donor parent are selected and repeatedly crossed, i.e., backcrossed, to the recurrent parent. The resulting plant is expected to have the attributes of the recurrent parent (i.e., variety) and the desirable trait transferred from the donor parent.

The single-seed descent procedure in the strict sense refers to planting a segregating population, harvesting a sample of one seed per plant, and using the one-seed sample to plant the next generation. When the population has been advanced from the F_2 to the desired level of inbreeding, the plants from which the lines are derived will each trace to different F_2 individuals. The number of plants in a population declines each generation due to failure of some seeds to germinate or some plants to produce at least one seed. As a result, not all of the F_2 plants originally sampled in the population will be represented by a progeny when generation advance is completed.

In a multiple-seed procedure, soybean breeders commonly harvest one or more pods from each plant in a population and thresh them together to form a bulk. Part of the bulk is used to plant the next generation and part is put in reserve. This procedure is also referred to as modified single-seed descent or the pod-bulk technique.

The multiple-seed procedure has been used to save labor at harvest. It is considerably faster to thresh pods with a machine than to remove one seed from each by hand as is required for the single-seed procedure. The multiple-seed procedure also makes it possible to plant the same number of seeds of a population each generation of inbreeding. Enough seeds are harvested to make up for those plants that did not germinate or produce seed.

Descriptions of other breeding methods that are commonly used for different traits and crops can be found in one of several reference books (e.g., Allard, "Principles of Plant Breeding," John Wiley & Sons, NY, University of California, Davis, California, 50-98, 1960; Simmonds, "Principles of Crop Improvement," Longman, Inc., NY, 369-399, 1979; Sneepe et al., "Plant breeding perspectives," Wageningen (ed), Centre for Agricultural Publishing and Documentation, 1979; Fehr, In: "Soybeans: Improvement, Production and Uses," 2d Ed., *Monograph* 16:249, 1987; Fehr, "Principles of Cultivar Development," Theory and Technique (Vol 1) and Crop Species Soybean (Vol 2), Iowa State Univ., Macmillan Pub. Co., NY, 360-376, 1987; Poehlman and Sleper, "Breeding Field Crops", 4th Ed., Iowa State University Press, Ames, 1995; Sprague and Dudley, eds., *Corn and Improvement*, 5th ed., 2006).

Proper testing should detect any major faults and establish the level of superiority or improvement over current varieties. In addition to showing superior performance, there must be a demand for a new variety that is compatible with industry standards or which creates a new market. The introduction of a new variety will incur additional costs to the seed producer, the grower, processor, and consumer due in part to special advertising and marketing, altered seed and commercial production practices, and new product utilization. The testing preceding release of a new variety should take into consideration research and development costs as well as the technical superiority of the final variety. For seed-propagated varieties, it must be feasible to produce seed easily and economically.

In addition to phenotypic observations, a plant can also be identified by its genotype. The genotype of a plant can be characterized through a molecular marker profile, which can identify plants of the same variety or a related variety, can identify plants and plant parts which are genetically superior as a result of an event comprising a backcross conversion, transgene, or genetic sterility factor, or can be used to determine or validate a pedigree. Such molecular marker profiling can be accomplished using a variety of techniques including, but not limited to, restriction fragment length polymorphism (RFLP), amplified fragment length polymor-

phism (AFLP), sequence-tagged sites (STS), randomly amplified polymorphic DNA (RAPD), diversity array technology (DART), arbitrarily primed polymerase chain reaction (AP-PCR), DNA amplification fingerprinting (DAF), sequence characterized amplified regions (SCARs), variable number tandem repeat (VNTR), short tandem repeat (STR), single feature polymorphism (SFP), simple sequence length polymorphism (SSLP), restriction site associated DNA, allozymes, isozyme markers, single nucleotide polymorphisms (SNPs), or simple sequence repeat (SSR) markers, also known as microsatellites (Gupta et al., 1999; Korzun et al., 2001). Various types of these markers, for example, can be used to identify individual plant varieties developed from specific parent varieties, as well as cells, seeds or other plant parts thereof. For example, see Cregan et al. (1999) "An Integrated Genetic Linkage Map of the Soybean Genome" *Crop Science* 39:1464-1490, and Berry et al. (2003) "Assessing Probability of Ancestry Using Simple Sequence Repeat Profiles: Applications to Maize Inbred Lines and Soybean Varieties" *Genetics* 165(1):331-342, each of which are incorporated by reference herein in their entirety.

In some examples, one or more markers may be used to examine, characterize and/or evaluate a soybean variety. Particular markers used for these purposes are not limited to any particular set of markers, but are envisioned to include any type of marker and marker profile that provides a means for distinguishing varieties. One method of genetic comparison may be to use only homozygous loci in the genome of the soybean variety 01098164.

Primers and PCR protocols for assaying these and other markers are disclosed in, for example, Soybase (sponsored by the USDA Agricultural Research Service and Iowa State University) located on the world wide web at 129.186.26/94/SSR.html. In addition to being used for identification of soybean variety 01098164, as well as plant parts and plant cells of soybean variety 01098164, a genetic profile may be used to identify a soybean plant produced through the use of soybean variety 01098164 or to verify a pedigree for progeny plants produced through the use of soybean variety 01098164. A genetic marker profile may also be useful in breeding and developing backcross conversions.

In an embodiment, the present invention provides a soybean plant characterized by molecular data obtained from a representative sample of said variety deposited with the Provasoli-Guillard National Center for Marine Algae and Microbiota (NCMA). Thus, plants, seeds, or parts thereof, having all or essentially all of the morphological and physiological characteristics of soybean variety 01098164 are provided. Further provided is a soybean plant formed by the combination of the disclosed soybean plant or plant cell with another soybean plant or cell and comprising the homozygous alleles of the variety.

In some examples, a plant, a plant part, or a seed of soybean variety 01098164 may be characterized by producing a molecular profile. A molecular profile may include, but is not limited to, one or more genotypic and/or phenotypic profile(s). A genotypic profile may include, but is not limited to, a marker profile, such as a genetic map, a linkage map, a trait maker profile, a SNP profile, an SSR profile, a genome-wide marker profile, a haplotype, and the like. A molecular profile may also be a nucleic acid sequence profile, and/or a physical map. A phenotypic profile may include, but is not limited to, a protein expression profile, a metabolic profile, an mRNA expression profile, and the like.

One means of performing genetic marker profiles is using SSR polymorphisms that are well known in the art. A marker system based on SSRs can be highly informative in linkage

analysis relative to other marker systems, in that multiple alleles may be present. Another advantage of this type of marker is that through use of flanking primers, detection of SSRs can be achieved, for example, by using the polymerase chain reaction (PCR), thereby eliminating the need for labor-intensive Southern hybridization. PCR detection may be performed using two oligonucleotide primers flanking the polymorphic segment of repetitive DNA to amplify the SSR region.

Following amplification, markers can be scored by electrophoresis of the amplification products. Scoring of marker genotype is based on the size of the amplified fragment, which correlates to the number of base pairs of the fragment. While variation in the primer used or in the laboratory procedures can affect the reported fragment size, relative values should remain constant regardless of specific primer or laboratory used. When comparing varieties, it may be beneficial to have all profiles performed in the same lab. Primers that can be used are publicly available and may be found in, for example, Soybase or Cregan et al. (*Crop Science* 39:1464-1490, 1999).

A genotypic profile of soybean variety 01098164 can be used to identify a plant comprising variety 01098164 as a parent, since such plants will comprise the same homozygous alleles as variety 01098164. Because the soybean variety is essentially homozygous at all relevant loci, most loci should have only one type of allele present. In contrast, a genetic marker profile of an F_1 progeny should be the sum of those parents, e.g., if one parent was homozygous for allele X at a particular locus, and the other parent homozygous for allele Y at that locus, then the F_1 progeny will be XY (heterozygous) at that locus. Subsequent generations of progeny produced by selection and breeding are expected to be of genotype XX (homozygous), YY (homozygous), or XY (heterozygous) for that locus position. When the F_1 plant is selfed or sibbed for successive filial generations, the locus should be either X or Y for that position.

In addition, plants and plant parts substantially benefiting from the use of variety 01098164 in their development, such as variety 01098164 comprising a backcross conversion, transgene, or genetic sterility factor, may be identified by having a molecular marker profile with a high percent identity to soybean variety 01098164. Such a percent identity might be 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5% or 99.9% identical to soybean variety 01098164.

A genotypic profile of variety 01098164 also can be used to identify essentially derived varieties and other progeny varieties developed from the use of variety 01098164, as well as cells and other plant parts thereof. Plants of the invention include any plant having at least 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 99.9% of the markers in the genotypic profile, and that retain 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 99.9% of the morphological and physiological characteristics of variety 01098164 when grown under the same conditions. Such plants may be developed using markers well known in the art. Progeny plants and plant parts produced using variety 01098164 may be identified, for example, by having a molecular marker profile of at least 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 99.5% genetic contribution from soybean variety 01098164, as measured by either percent identity or percent similarity. Such progeny may be further characterized as being within a pedigree distance of

variety 01098164, such as within 1, 2, 3, 4, or 5 or less cross pollinations to a soybean plant other than variety 01098164, or a plant that has variety 01098164 as a progenitor. Unique molecular profiles may be identified with other molecular tools, such as SNPs and RFLPs.

Any time the soybean variety 01098164 is crossed with another, different, variety, first generation (F₁) soybean progeny are produced. The hybrid progeny are produced regardless of characteristics of the two varieties produced. As such, an F₁ hybrid soybean plant may be produced by crossing 01098164 with any second soybean plant. The second soybean plant may be genetically homogeneous (e.g., inbred) or may itself be a hybrid. Therefore, any F₁ hybrid soybean plant produced by crossing soybean variety 01098164 with a second soybean plant is a part of the present invention.

Soybean plants (*Glycine max* L.) can be crossed by either natural or mechanical techniques (see, e.g., Fehr, "Soybean," In: *Hybridization of Crop Plants*, Fehr and Hadley (eds), *Am. Soc. Agron. and Crop Sci. Soc. Am.*, Madison, WI, 590-599, 1980). Natural pollination occurs in soybeans either by self-pollination or natural cross-pollination, which typically is aided by pollinating organisms. In either natural or artificial crosses, flowering and flowering time are an important consideration. Soybean is a short-day plant, but there is considerable genetic variation for sensitivity to photoperiod (Hamner, "*Glycine max*(L.) Merrill," In: *The Induction of Flowering: Some Case Histories*, Evans (ed), Cornell Univ. Press, Ithaca, NY, 62-89, 1969; Criswell and Hume, *Crop Sci.*, 12:657-660, 1972). The critical day length for flowering ranges from about 13 h for genotypes adapted to tropical latitudes to 24 h for photoperiod-insensitive genotypes grown at higher latitudes (Shibles et al., "Soybean," In: *Crop Physiology: Some Case Histories*, Evans (ed), Cambridge Univ. Press, Cambridge, England, 51-189, 1975). Soybeans seem to be insensitive to day length for 9 days after emergence. Photoperiods shorter than the critical day length are required for 7 to 26 days to complete flower induction (Borthwick and Parker, *Bot. Gaz.*, 100:374-387, 1938; Shanmugasundaram and Tsou, *Crop Sci.*, 18:598-601, 1978).

Sensitivity to day length is an important consideration when genotypes are grown outside of their area of adaptation. When genotypes adapted to tropical latitudes are grown in the field at higher latitudes, they may not mature before frost occurs. Plants can be induced to flower and mature earlier by creating artificially short days or by grafting (Fehr, "Soybean," In: *Hybridization of Crop Plants*, Fehr and Hadley (eds), *Am. Soc. Agron. and Crop Sci. Soc. Am.*, Madison, WI, 590-599, 1980). Soybeans frequently are grown in winter nurseries located at sea level in tropical latitudes where day lengths are much shorter than their critical photoperiod. The short day lengths and warm temperatures encourage early flowering and seed maturation, and genotypes can produce a seed crop in 90 days or fewer after planting. Early flowering is useful for generation advance when only a few self-pollinated seeds per plant are needed, but not for artificial hybridization because the flowers self-pollinate before they are large enough to manipulate for hybridization. Artificial lighting can be used to extend the natural day length to about 14.5 h to obtain flowers suitable for hybridization and to increase yields of self-pollinated seed.

The effect of a short photoperiod on flowering and seed yield can be partly offset by altitude, probably due to the effects of cool temperature (Major et al., *Crop Sci.*, 15:174-179, 1975). At tropical latitudes, varieties adapted to the

northern U.S. perform more like those adapted to the southern U.S. at high altitudes than they do at sea level.

The light level required to delay flowering is dependent on the quality of light emitted from the source and the genotype being grown. Blue light with a wavelength of about 480 nm requires more than 30 times the energy to inhibit flowering as red light with a wavelength of about 640 nm (Parker et al., *Bot. Gaz.*, 108:1-26, 1946).

Temperature can also play a significant role in the flowering and development of soybean plants (Major et al., *Crop Sci.*, 15:174-179, 1975). It can influence the time of flowering and suitability of flowers for hybridization. Temperatures below 21° C. or above 32° C. can reduce floral initiation or seed set (Hamner, "*Glycine max*(L.) Merrill," In: *The Induction of Flowering: Some Case Histories*, Evans (ed), Cornell Univ. Press, Ithaca, NY, 62-89, 1969; van Schaik and Probst, *Agron. J.*, 50:192-197, 1958). Artificial hybridization is most successful between 26° C. and 32° C. because cooler temperatures reduce pollen shed and result in flowers that self-pollinate before they are large enough to manipulate. Warmer temperatures frequently are associated with increased flower abortion caused by moisture stress; however, successful crosses are possible at about 35° C. if soil moisture is adequate.

Soybeans have been classified as indeterminate, semi-determinate, and determinate based on the abruptness of stem termination after flowering begins (Bernard and Weiss, "Qualitative genetics," In: *Soybeans: Improvement, Production, and Uses*, Caldwell (ed), *Am. Soc. of Agron.*, Madison, WI, 117-154, 1973). When grown at their latitude of adaptation, indeterminate genotypes flower when about one-half of the nodes on the main stem have developed. They have short racemes with few flowers, and their terminal node has only a few flowers. Semi-determinate genotypes also flower when about one-half of the nodes on the main stem have developed, but node development and flowering on the main stem stops more abruptly than on indeterminate genotypes. Their racemes are short and have few flowers, except for the terminal one, which may have several times more flowers than those lower on the plant. Determinate varieties begin flowering when all or most of the nodes on the main stem have developed. They usually have elongated racemes that may be several centimeters in length and may have a large number of flowers. Stem termination and flowering habit are reported to be controlled by two major genes (Bernard and Weiss, "Qualitative genetics," In: *Soybeans: Improvement, Production, and Uses*, Caldwell (ed), *Am. Soc. of Agron.*, Madison, WI, 117-154, 1973).

Soybean flowers typically are self-pollinated on the day the corolla opens. The amount of natural crossing, which is typically associated with insect vectors such as honeybees, is approximately 1% for adjacent plants within a row and 0.5% between plants in adjacent rows (Boerma and Moradshahi, *Crop Sci.*, 15:858-861, 1975). The structure of soybean flowers is similar to that of other legume species and consists of a calyx with five sepals, a corolla with five petals, 10 stamens, and a pistil (Carlson, "Morphology," In: *Soybeans: Improvement, Production, and Uses*, Caldwell (ed), *Am. Soc. of Agron.*, Madison, WI, 17-95, 1973). The calyx encloses the corolla until the day before anthesis. The corolla emerges and unfolds to expose a standard, two wing petals, and two keel petals. An open flower is about 7 mm long from the base of the calyx to the tip of the standard and 6 mm wide across the standard. The pistil consists of a single ovary that contains one to five ovules, a style that curves toward the standard, and a club-shaped stigma. The stigma is receptive to pollen about 1 day before anthesis and

remains receptive for 2 days after anthesis, if the flower petals are not removed. Filaments of nine stamens are fused, and the one nearest the standard is free. The stamens form a ring below the stigma until about 1 day before anthesis, then their filaments begin to elongate rapidly and elevate the anthers around the stigma. The anthers dehisce on the day of anthesis, pollen grains fall on the stigma, and within 10 h the pollen tubes reach the ovary and fertilization is completed (Johnson and Bernard, "Soybean genetics and breeding," *In: The Soybean*, Norman (ed), Academic Press, NY, 1-73, 1963).

Self-pollination occurs naturally in soybean with no manipulation of the flowers. For the crossing of two soybean plants, it is often beneficial, although not required, to utilize artificial hybridization. In artificial hybridization, the flower used as a female in a cross is manually cross pollinated prior to maturation of pollen from the flower, thereby preventing self-fertilization, or alternatively, the male parts of the flower are emasculated using a technique known in the art. Techniques for emasculating the male parts of a soybean flower include, for example, physical removal of the male parts, use of a genetic factor conferring male sterility, and application of a chemical gametocide to the male parts.

For artificial hybridization employing emasculation, flowers that are expected to open the following day are selected on the female parent. The buds are swollen and the corolla is just visible through the calyx or has begun to emerge. The selected buds on a parent plant are prepared and all of the self-pollinated flowers or immature buds are removed. Special care is required to remove immature buds that are hidden under the stipules at the leaf axil, which could develop into flowers at a later date. To remove a flower, the flower is grasped and the location of the stigma is determined by examining the sepals. A long, curvy sepal covers the keel, and the stigma is on the opposite side of the flower. The calyx is removed by pulling each sepal down and around the flower. The exposed corolla is then removed just above the calyx scar, taking care to remove the keel petals without injuring the stigma. The ring of anthers is visible after the corolla is removed, unless the anthers were removed with the petals. Cross-pollination can then be carried out using, for example, petri dishes or envelopes in which male flowers have been collected. Desiccators containing calcium chloride crystals are used in some environments to dry male flowers to obtain adequate pollen shed.

It has been demonstrated that emasculation is unnecessary to prevent self-pollination (Walker et al., *Crop Sci.*, 19:285-286, 1979). When emasculation is not used, the anthers near the stigma frequently are removed to make it clearly visible for pollination. The female flower usually is hand-pollinated immediately after it is prepared; although a delay of several hours does not seem to reduce seed set. Pollen shed typically begins in the morning and may end when temperatures are above 30° C., or may begin later and continue throughout much of the day with more moderate temperatures.

Pollen is available from a flower with a recently opened corolla, but the degree of corolla opening associated with pollen shed may vary during the day. In many environments, it is possible to collect male flowers and use them immediately without storage. In the southern U.S. and other humid climates, pollen shed occurs in the morning when female flowers are more immature and difficult to manipulate than in the afternoon, and the flowers may be damp from heavy dew. In those circumstances, male flowers may be collected into envelopes or petri dishes in the morning and the open container placed in a desiccator for about 4 h at a temperature of about 25° C. The desiccator may be taken to the field

in the afternoon and kept in the shade to prevent excessive temperatures from developing within it. Pollen viability can be maintained in flowers for up to 2 days when stored at about 5° C. In a desiccator at 3° C., flowers can be stored successfully for several weeks; however, varieties may differ in the percentage of pollen that germinates after long-term storage (Kuehl, "Pollen viability and stigma receptivity of *Glycine max* (L.) Merrill," Thesis, North Carolina State College, Raleigh, N C, 1961).

Either with or without emasculation of the female flower, hand pollination can be carried out by removing the stamens and pistil with a forceps from a flower of the male parent and gently brushing the anthers against the stigma of the female flower. Access to the stamens can be achieved by removing the front sepal and keel petals or piercing the keel with closed forceps and allowing them to open to push the petals away. Brushing the anthers on the stigma causes them to rupture, and the highest percentage of successful crosses is obtained when pollen is clearly visible on the stigma. Pollen shed can be checked by tapping the anthers before brushing the stigma. Several male flowers may have to be used to obtain suitable pollen shed when conditions are unfavorable or the same male with good pollen shed may be used to pollinate several flowers.

Alternatively, liquid-mediated delivery of pollen to an enclosed stigma of a plant can facilitate higher throughput and improved efficiency of cross-pollination. See, for example, U.S. Patent Application Publication No. 20210092920, the disclosure of which is specifically incorporated herein by reference in its entirety. An "enclosed stigma" refers to a stigma on a flower that is still enclosed by structures such as petals and sepals. To ensure a successful cross-pollination using the liquid-mediated pollen delivery system, recipient female flowers must be used at the optimal developmental stage. At this stage, the stigma is receptive to pollen grains and the anthers have not yet shed pollen.

Purified pollen grains from a donor plant, obtained by methods known in the art, may be suspended in any suitable solution, including, for example, a sucrose solution or oil, and stored at low temperature to preserve pollen viability. The density of the pollen grains in the solution should be sufficient to provide successful pollination using an appropriate volume of the solution. The pollen solution can then be introduced into a flower bud of a recipient plant by injecting the solution into the bud such that the solution contacts the stigma. Injecting may be performed using any instrument having the capacity to inject a desired amount of solution in the flower bud without disrupting the stigma and ovule (for example, a syringe).

When male flowers do not have to be collected and dried in a desiccator, it may be desired to plant the parents of a cross adjacent to each other. Plants usually are grown in rows 65 to 100 cm apart to facilitate movement of personnel within the field nursery. Yield of self-pollinated seed from an individual plant may range from a few seeds to more than 1,000 as a function of plant density. A density of 30 plants/m of row can be used when 30 or fewer seeds per plant is adequate, 10 plants/m can be used to obtain about 100 seeds/plant, and 3 plants/m usually results in maximum seed production per plant. Densities of 12 plants/m or less commonly are used for artificial hybridization.

Multiple planting dates about 7 to 14 days apart usually are used to match parents of different flowering dates. When differences in flowering dates are extreme between parents, flowering of the later parent can be hastened by creating an artificially short day or flowering of the earlier parent can be

delayed by use of artificially long days or delayed planting. For example, crosses with genotypes adapted to the southern U.S. are made in northern U.S. locations by covering the late genotype with a box, large can, or similar container to create an artificially short photoperiod of about 12 h for about 15 days beginning when there are three nodes with trifoliate leaves on the main stem. Plants induced to flower early tend to have flowers that self-pollinate when they are small and can be difficult to prepare for hybridization.

Grafting can be used to hasten the flowering of late flowering genotypes. A scion from a late genotype grafted on a stock that has begun to flower will begin to bloom up to 42 days earlier than normal (Kiihl et al., *Crop Sci.*, 17:181-182, 1977). First flowers on the scion appear from 21 to 50 days after the graft.

Observing pod development 7 days after pollination generally is adequate to identify a successful cross. Abortion of pods and seeds can occur several weeks after pollination, but the percentage of abortion usually is low if plant stress is minimized (Shibles et al., "Soybean," In: *Crop Physiology: Some Case Histories*, Evans (ed), Cambridge Univ. Press, Cambridge, England, 51-189, 1975). Pods that develop from artificial hybridization can be distinguished from self-pollinated pods by the presence of the calyx scar, caused by removal of the sepals. The sepals begin to fall off as the pods mature; therefore, harvest should be completed at or immediately before the time the pods reach their mature color. Harvesting pods early also avoids any loss by shattering.

Once harvested, pods are typically air-dried at not more than 38° C. until the seeds contain 13% moisture or less, then the seeds are removed by hand. Seed can be stored satisfactorily at about 25° C. for up to a year if relative humidity is 50% or less. In humid climates, germination percentage declines rapidly unless the seed is dried to 7% moisture and stored in an air-tight container at room temperature. Long-term storage in any climate is best accomplished by drying seed to 7% moisture and storing it at 10° C. or less in a room maintained at 50% relative humidity or in an air-tight container.

Further Embodiments of the Invention

In certain aspects of the invention, plants of soybean variety 01098164 are modified to include at least a first heritable trait. Such plants may, in one embodiment, be developed by a plant breeding technique called backcrossing, wherein essentially all of the morphological and physiological characteristics of a variety are recovered in addition to a genetic locus transferred into the plant via the backcrossing technique. By essentially all of the morphological and physiological characteristics, it is meant that the characteristics of a plant are recovered that are otherwise present when compared in the same environment, other than occasional variant traits that might arise during backcrossing or direct introduction of a transgene. It is understood that a locus introduced by backcrossing may or may not be transgenic in origin, and thus the term backcrossing specifically includes backcrossing to introduce loci that were created by genetic transformation.

In a typical backcross protocol, the original variety of interest (recurrent parent) is crossed to a second variety (nonrecurrent parent) that carries the single locus of interest to be transferred. The resulting progeny from this cross are then crossed again to the recurrent parent and the process is repeated until a soybean plant is obtained wherein essentially all of the morphological and physiological character-

istics of the recurrent parent are recovered in the converted plant, in addition to the transferred locus from the nonrecurrent parent.

The selection of a suitable recurrent parent is an important step for a successful backcrossing procedure. The goal of a backcross protocol is to alter or substitute a trait or characteristic in the original variety. To accomplish this, a locus of the recurrent variety is modified or substituted with the desired locus from the nonrecurrent parent, while retaining essentially all of the rest of the genome of the original variety, and therefore the morphological and physiological constitution of the original variety. The choice of the particular nonrecurrent parent will depend on the purpose of the backcross; one of the major purposes is to add some commercially or agronomically important trait to the plant. The exact backcrossing protocol will depend on the characteristic or trait being altered to determine an appropriate testing protocol. Although backcrossing methods are simplified when the characteristic being transferred is a dominant allele, a recessive allele may also be transferred. In this instance, it may be necessary to introduce a test of the progeny to determine if the desired characteristic has been successfully transferred.

Soybean varieties can also be developed from more than two parents (Fehr, In: "*Soybeans: Improvement, Production and Uses*," 2nd Ed., *Monograph* 16:249, 1987). The technique, known as modified backcrossing, uses different recurrent parents during the backcrossing. Modified backcrossing may be used to replace the original recurrent parent with a variety having certain more desirable characteristics or multiple parents may be used to obtain different desirable characteristics from each.

Many traits have been identified that are not regularly selected for in the development of a new inbred but that can be improved by backcrossing techniques. Traits may or may not be transgenic; examples of these traits include, but are not limited to, male sterility, herbicide resistance, resistance to bacterial, fungal, or viral disease, insect and pest resistance, restoration of male fertility, enhanced nutritional quality, yield stability, and yield enhancement. These comprise genes generally inherited through the nucleus.

Direct selection may be applied when the locus acts as a dominant trait. An example of a dominant trait is the herbicide resistance trait. For this selection process, the progeny of the initial cross are sprayed with the herbicide prior to the backcrossing. The spraying eliminates any plants which do not have the desired herbicide resistance characteristic, and only those plants that have the herbicide resistance gene are used in the subsequent backcross. This process is then repeated for all additional backcross generations.

Selection of soybean plants for breeding is not necessarily dependent on the phenotype of a plant and instead can be based on genetic investigations. For example, one may utilize a suitable genetic marker that is closely associated with a trait of interest. One of these markers may therefore be used to identify the presence or absence of a trait in the offspring of a particular cross, and hence may be used in selection of progeny for continued breeding. This technique may commonly be referred to as marker assisted selection. Any other type of genetic marker or other assay that is able to identify the relative presence or absence of a trait of interest in a plant may also be useful for breeding purposes. Procedures for marker assisted selection applicable to the breeding of soybeans are well known in the art. Such methods will be of particular utility in the case of recessive traits and variable phenotypes, or when conventional assays

may be more expensive, time consuming or otherwise disadvantageous. Genetic markers that could be used in accordance with the invention include, but are not necessarily limited to, Simple Sequence Length Polymorphisms (SSLPs) (Williams et al., *Nucleic Acids Res.*, 18:6531-6535, 1990), Randomly Amplified Polymorphic DNAs (RAPDs), DNA Amplification Fingerprinting (DAF), Sequence Characterized Amplified Regions (SCARs), Arbitrary Primed Polymerase Chain Reaction (AP-PCR), Amplified Fragment Length Polymorphisms (AFLPs) (European Patent Application Publication No. EP0534858, specifically incorporated herein by reference in its entirety), and Single Nucleotide Polymorphisms (SNPs) (Wang et al., *Science*, 280:1077-1082, 1998).

Many qualitative characters also have a potential use as phenotype-based genetic markers in soybeans; however, some or many may not differ among varieties commonly used as parents (Bernard and Weiss, "Qualitative genetics," *In: Soybeans: Improvement, Production, and Uses*, Caldwell (ed), *Am. Soc. of Agron.*, Madison, WI, 117-154, 1973). The most widely used genetic markers are flower color (purple dominant to white), pubescence color (brown dominant to gray), and pod color (brown dominant to tan). The association of purple hypocotyl color with purple flowers and green hypocotyl color with white flowers is commonly used to identify hybrids in the seedling stage. Differences in maturity, height, hilum color, and pest resistance between parents can also be used to verify hybrid plants.

Many useful traits that can be introduced by backcrossing, as well as directly into a plant, are those that are introduced by genetic transformation techniques. Genetic transformation may therefore be used to insert a selected transgene into the soybean variety of the invention or may, alternatively, be used for the preparation of transgenes which can be introduced by backcrossing. Methods for the transformation of many economically important plants, including soybeans, are well known to those of skill in the art. Techniques which may be employed to genetically transform soybeans include, but are not limited to, electroporation, microprojectile bombardment, *Agrobacterium*-mediated transformation and direct DNA uptake by protoplasts.

To effect transformation by electroporation, one may employ either friable tissues, such as a suspension culture of cells or embryogenic callus or alternatively one may transform immature embryos or other organized tissue directly. In this technique, one would partially degrade the cell walls of the chosen cells by exposing them to pectin-degrading enzymes (pectolyases) or mechanically wound tissues in a controlled manner.

Protoplasts may also be employed for electroporation transformation of plants (Bates, *Mol. Biotechnol.*, 2(2):135-145, 1994; Lazzeri, *Methods Mol. Biol.*, 49:95-106, 1995). For example, the generation of transgenic soybean plants by electroporation of cotyledon-derived protoplasts was described by Dhir and Widholm in International Patent Application Publication No. WO 92/17598, the disclosure of which is specifically incorporated herein by reference.

An efficient method for delivering transforming DNA segments to plant cells is microprojectile bombardment. In this method, particles are coated with nucleic acids and delivered into cells by a propelling force. Exemplary particles include those comprised of tungsten, platinum, or gold. For the bombardment, cells in suspension are concentrated on filters or solid culture medium. Alternatively, immature embryos or other target cells may be arranged on

solid culture medium. The cells to be bombarded are positioned at an appropriate distance below the macroprojectile stopping plate.

An illustrative embodiment of a method for delivering DNA into plant cells by acceleration is the Biolistics Particle Delivery System, which can be used to propel particles coated with DNA or cells through a screen, such as a stainless steel or Nytex screen, onto a surface covered with target soybean cells. The screen disperses the particles so that they are not delivered to the recipient cells in large aggregates. It is believed that a screen intervening between the projectile apparatus and the cells to be bombarded reduces the size of the projectile aggregate and may contribute to a higher frequency of transformation by reducing the damage inflicted on the recipient cells by projectiles that are too large.

Microprojectile bombardment techniques are widely applicable, and may be used to transform virtually any plant species. The application of microprojectile bombardment for the transformation of soybeans is described, for example, in U.S. Pat. No. 5,322,783, the disclosure of which is specifically incorporated herein by reference in its entirety.

Agrobacterium-mediated transfer is another widely applicable system for introducing gene loci into plant cells. An advantage of the technique is that DNA can be introduced into whole plant tissues, thereby bypassing the need for regeneration of an intact plant from a protoplast. Modern *Agrobacterium* transformation vectors are capable of replication in *E. coli* as well as *Agrobacterium*, allowing for convenient manipulations (Klee et al., *Bio. Tech.*, 3(7):637-642, 1985). Moreover, recent technological advances in vectors for *Agrobacterium*-mediated gene transfer have improved the arrangement of genes and cloning sites in the vectors to facilitate the construction of vectors capable of expressing various polypeptide coding genes. Vectors can have convenient multiple-cloning sites (MCS) flanked by a promoter and a polyadenylation site for direct expression of inserted polypeptide coding genes. Other vectors can comprise site-specific recombination sequences, enabling insertion of a desired DNA sequence without the use of restriction enzymes (Curtis et al., *Plant Physiology* 133:462-469, 2003). Additionally, *Agrobacterium* containing both armed and disarmed Ti genes can be used for transformation.

In those plant strains in which *Agrobacterium*-mediated transformation is efficient, it is the method of choice because of the facile and defined nature of the gene locus transfer. The use of *Agrobacterium*-mediated plant integrating vectors to introduce DNA into plant cells is well known in the art (Fraley et al., *Bio. Tech.*, 3(7):629-635, 1985; U.S. Pat. No. 5,563,055). Use of *Agrobacterium* in the context of soybean transformation has been described, for example, by Chee and Slightom (*Methods Mol. Biol.*, 44:101-119, 1995) and in U.S. Pat. No. 5,569,834, the disclosures of which are specifically incorporated herein by reference in their entirety.

Transformation of plant protoplasts also can be achieved using methods based on calcium phosphate precipitation, polyethylene glycol treatment, electroporation, and combinations of these treatments (see, e.g., Potrykus et al., *Mol. Gen. Genet.*, 199(2):169-177, 1985; Omirulleh et al., *Plant Mol. Biol.*, 21(3):415-428, 1993; Fromm et al., *Nature*, 319(6056):791-793, 1986; Uchimiya et al., *Mol. Gen. Genet.*, 204(2):204-207, 1986; Marcotte et al., *Nature*, 335(6189):454-457, 1988). The demonstrated ability to regenerate soybean plants from protoplasts makes each of these techniques applicable to soybean (Dhir et al., *Plant Cell Rep.*, 10(2):97-101, 1991).

Included among various plant transformation techniques are methods permitting the site-specific modification of a plant genome. These modifications can include, but are not limited to, site-specific mutations, deletions, insertions, and replacements of nucleotides. These modifications can be made anywhere within the genome of a plant, for example, in genomic elements, including, among others, coding sequences, regulatory elements, and non-coding DNA sequences. Any number of such modifications can be made and that number of modifications may be made in any order or combination, for example, simultaneously all together or one after another. Such methods may be used to modify a particular trait conferred by a locus. The techniques for making such modifications by genome editing are well known in the art and include, for example, use of CRISPR-Cas systems, zinc-finger nucleases (ZFNs), and transcription activator-like effector nucleases (TALENs), among others.

Many hundreds if not thousands of different genes are known and could potentially be introduced into a soybean plant according to the invention. Non-limiting examples of particular genes and corresponding phenotypes one may choose to introduce into a soybean plant are presented below.

A. Herbicide Resistance

Numerous herbicide resistance genes are known and may be employed with the invention. A non-limiting example is a gene conferring resistance to a herbicide that inhibits the growing point or meristem such as imidazolinone or sulfonyleurea herbicides. As imidazolinone and sulfonyleurea herbicides are acetolactate synthase (ALS)-inhibiting herbicides that prevent the formation of branched chain amino acids, exemplary genes in this category code for ALS and AHAS enzymes as described, for example, by Lee et al., *EMBO J.*, 7:1241, 1988; Gleen et al., *Plant Molec. Biology*, 18:1185, 1992; and Miki et al., *Theor. Appl. Genet.*, 80:449, 1990. As a non-limiting example, a gene may be employed to confer resistance to the exemplary sulfonyleurea herbicide nicosulfuron.

Resistance genes for glyphosate (resistance conferred by mutant 5-enolpyruvylshikimate-3-phosphate synthase (EPSPS) and *aroA* genes, respectively) and other phosphono compounds such as glufosinate (phosphinothricin acetyltransferase (PAT) and *Streptomyces hygroscopicus* phosphinothricin acetyltransferase (*bar*) genes) may also be used. See, for example, U.S. Pat. No. 4,940,835 to Shah et al., which discloses the nucleotide sequence of a form of EPSPS that can confer glyphosate resistance. Non-limiting examples of EPSPS transformation events conferring glyphosate resistance are provided by U.S. Pat. Nos. 6,040,497 and 7,632,985. The MON89788 event disclosed in U.S. Pat. No. 7,632,985 in particular is beneficial in conferring glyphosate tolerance in combination with an increase in average yield relative to prior events.

A DNA molecule encoding a mutant *aroA* gene can be obtained under ATCC Accession No. 39256, and the nucleotide sequence of the mutant gene is disclosed in U.S. Pat. No. 4,769,061 to Comai. A hygromycin B phosphotransferase gene from *E. coli* that confers resistance to glyphosate in tobacco callus and plants is described in Penaloza-Vazquez et al., *Plant Cell Reports*, 14:482, 1995. European Patent Application Publication No. EP0333033 to Kumada et al., and U.S. Pat. No. 4,975,374 to Goodman et al., disclose nucleotide sequences of glutamine synthetase genes that confer resistance to herbicides such as L-phosphinothricin. The nucleotide sequence of a phosphinothricin acetyltransferase gene is provided in European Patent Application Publication No. EP0242246 to Leemans et al. DeGreef et al.

(*Biotechnology*, 7:61, 1989) describe the production of transgenic plants that express chimeric *bar* genes coding for phosphinothricin acetyl transferase activity. Exemplary genes conferring resistance to a phenoxy class herbicide haloxyfop and a cyclohexanedione class herbicide sethoxydim are the *Acct-S1*, *Acct-S2* and *Acct-S3* genes described by Marshall et al., (*Theor. Appl. Genet.*, 83:435, 1992). As a non-limiting example, a gene may confer resistance to other exemplary phenoxy class herbicides that include, but are not limited to, quizalofop-p-ethyl and 2,4-dichlorophenoxyacetic acid (2,4-D).

Genes are also known that confer resistance to herbicides that inhibit photosynthesis such as, for example, triazine herbicides (*psbA* and *gs+* genes) and benzonitrile herbicides (nitrilase gene). As a non-limiting example, a gene may confer resistance to the exemplary benzonitrile herbicide bromoxynil. Przibila et al. (*Plant Cell*, 3:169, 1991) describe the transformation of *Chlamydomonas* with plasmids encoding mutant *psbA* genes. Nucleotide sequences for nitrilase genes are disclosed in U.S. Pat. No. 4,810,648 to Stalker, and DNA molecules containing these genes are available under ATCC Accession Nos. 53435, 67441, and 67442. Cloning and expression of DNA coding for a glutathione S-transferase is described by Hayes et al. (*Biochem. J.*, 285:173, 1992). 4-hydroxyphenylpyruvate dioxygenase (HPPD) is a target of the HPPD-inhibiting herbicides, which deplete plant plastoquinone and vitamin E pools. Rippert et al. (*Plant Physiol.*, 134:92, 2004) describes an HPPD-inhibitor resistant tobacco plant that was transformed with a yeast-derived prephenate dehydrogenase (PDH) gene. Protoporphyrinogen oxidase (PPO) is the target of the PPO-inhibitor class of herbicides; a PPO-inhibitor resistant PPO gene was recently identified in *Amaranthus tuberculatus* (Patzoldt et al., *PNAS*, 103(33):12329, 2006). The herbicide methyl viologen inhibits CO₂ assimilation. Foyer et al. (*Plant Physiol.*, 109:1047, 1995) describe a plant overexpressing glutathione reductase (GR) that is resistant to methyl viologen treatment.

Siminszky (*Phytochemistry Reviews*, 5:445, 2006) describes plant cytochrome P450-mediated detoxification of multiple, chemically unrelated classes of herbicides. Modified bacterial genes have been successfully demonstrated to confer resistance to atrazine, a herbicide that binds to the plastoquinone-binding membrane protein *Q_B* in photosystem II to inhibit electron transport. See, for example, studies by Cheung et al. (*PNAS*, 85:391, 1988), describing tobacco plants expressing the chloroplast *psbA* gene from an atrazine-resistant biotype of *Amaranthus hybridus* fused to the regulatory sequences of a nuclear gene, and Wang et al. (*Plant Biotech. J.*, 3:475, 2005), describing transgenic alfalfa, *Arabidopsis*, and tobacco plants expressing the *atzA* gene from *Pseudomonas* sp. that were able to detoxify atrazine.

Bayley et al. (*Theor. Appl. Genet.*, 83:645, 1992) describe the creation of 2,4-D-resistant transgenic tobacco and cotton plants using the 2,4-D monooxygenase gene *tfdA* from *Alcaligenes eutrophus* plasmid pJP5. U.S. Patent Application Publication No. 20030135879 describes the isolation of a gene for dicamba monooxygenase (DMO) from *Pseudomonas maltophilia* that is involved in the conversion of dicamba to a non-toxic 3,6-dichlorosalicylic acid and thus may be used for producing plants tolerant to this herbicide.

Other examples of herbicide resistance have been described, for instance, in U.S. Pat. Nos. 6,803,501; 6,448,476; 6,248,876; 6,225,114; 6,107,549; 5,866,775; 5,804,425; 5,633,435; 5,463,175.

B. Disease and Pest Resistance

Plant defenses are often activated by specific interaction between the product of a disease resistance gene (R) in the plant and the product of a corresponding avirulence (Avr) gene in the pathogen. A plant line can be transformed with a cloned resistance gene to engineer plants that are resistant to specific pathogen strains. See, for example Jones et al. (*Science*, 266:789-793, 1994) (cloning of the tomato Cf-9 gene for resistance to *Cladosporium fulvum*); Martin et al. (*Science*, 262:1432-1436, 1993) (tomato Pto gene for resistance to *Pseudomonas syringae* pv. tomato); and Mindrinos et al. (*Cell*, 78(6):1089-1099, 1994) (*Arabidopsis* RPS2 gene for resistance to *Pseudomonas syringae*).

A viral-invasive protein or a complex toxin derived therefrom may also be used for viral disease resistance. For example, the accumulation of viral coat proteins in transformed plant cells imparts resistance to viral infection and/or disease development effected by the virus from which the coat protein gene is derived and related viruses. See Beachy et al. (*Ann. Rev. Phytopathol.*, 28:451, 1990). Coat protein-mediated resistance has been conferred upon transformed plants against alfalfa mosaic virus, cucumber mosaic virus, tobacco streak virus, potato virus X, potato virus Y, tobacco etch virus, tobacco rattle virus, and tobacco mosaic virus.

A virus-specific antibody may also be used. See, for example, Tavladoraki et al. (*Nature*, 366:469-472, 1993), who show that transgenic plants expressing recombinant antibody genes are protected from virus attack. Virus resistance has also been described in, for example, U.S. Pat. Nos. 6,617,496; 6,608,241; 6,015,940; 6,013,864; 5,850,023 and 5,304,730. Additional means of inducing whole-plant resistance to a pathogen include modulation of the systemic acquired resistance (SAR) or pathogenesis related (PR) genes, for example genes homologous to the *Arabidopsis thaliana* NIM1/NPR1/SAI1, and/or by increasing salicylic acid production (Ryals et al., *Plant Cell*, 8:1809-1819, 1996).

Logemann et al. (*Biotechnology*, 10:305-308, 1992), for example, disclose transgenic plants expressing a barley ribosome-inactivating gene that have an increased resistance to fungal disease. Plant defensins may be used to provide resistance to fungal pathogens (Thomma et al., *Planta*, 216:193-202, 2002). Other examples of fungal disease resistance are provided in U.S. Pat. Nos. 6,653,280; 6,573,361; 6,506,962; 6,316,407; 6,215,048; 5,516,671; 5,773,696; 6,121,436; and 6,316,407.

Nematode resistance has been described in, for example, U.S. Pat. No. 6,228,992, and bacterial disease resistance has been described in, for example, U.S. Pat. No. 5,516,671.

The use of the herbicide glyphosate for disease control in soybean plants containing event MON89788, which confers glyphosate tolerance, has also been described in U.S. Pat. No. 7,608,761.

C. Insect Resistance

One example of an insect resistance gene includes a *Bacillus thuringiensis* protein, a derivative thereof, or a synthetic polypeptide modeled thereon. See, for example, Geiser et al. (*Gene*, 48(1):109-118, 1986), who disclose the cloning and nucleotide sequence of a *Bacillus thuringiensis* δ -endotoxin gene. Moreover, DNA molecules encoding δ -endotoxin genes can be purchased from the American Type Culture Collection, Manassas, Virginia, for example, under ATCC Accession Nos. 40098, 67136, 31995 and 31998. Another example is a lectin. See, for example, Van Damme et al., (*Plant Molec. Biol.*, 24:825-830, 1994), who disclose the nucleotide sequences of several *Clivia miniata* mannose-binding lectin genes. A vitamin-binding protein

may also be used, such as, for example, avidin. See PCT Application No. US93/06487, the contents of which are hereby incorporated by reference. This application teaches the use of avidin and avidin homologues as larvicides against insect pests.

Yet another insect resistance gene is an enzyme inhibitor, for example, protease, proteinase, or amylase inhibitors. See, for example, Abe et al. (*J. Biol. Chem.*, 262:16793-16797, 1987) describing the nucleotide sequence of a rice cysteine proteinase inhibitor; Linthorst et al. (*Plant Molec. Biol.*, 21:985-992, 1993) describing the nucleotide sequence of a cDNA encoding tobacco proteinase inhibitor I; and Sumitani et al. (*Biosci. Biotech. Biochem.*, 57:1243-1248, 1993) describing the nucleotide sequence of a *Streptomyces nitrosporeus* α -amylase inhibitor.

An insect-specific hormone or pheromone may also be used. See, for example, the disclosure by Hammock et al. (*Nature*, 344:458-461, 1990) of baculovirus expression of cloned juvenile hormone esterase, an inactivator of juvenile hormone; Gade and Goldsworthy (Eds. *Physiological System in Insects*, Elsevier Academic Press, Burlington, M A, 2007), describing allostatins and their potential use in pest control; and Palli et al. (*Vitam. Horm.*, 73:59-100, 2005), disclosing use of ecdysteroid and ecdysteroid receptor in agriculture. The diuretic hormone receptor (DHR) was identified in Price et al. (*Insect Mol. Biol.*, 13:469-480, 2004) as another potential candidate target of insecticides.

Still other examples include an insect-specific antibody or an immunotoxin derived therefrom and a developmental-arrestive protein. See Taylor et al. (Seventh Int'l Symposium on Molecular Plant-Microbe Interactions, Edinburgh, Scotland, Abstract W97, 1994), who described enzymatic inactivation in transgenic tobacco via production of single-chain antibody fragments. Numerous other examples of insect resistance have been described. See, for example, U.S. Pat. Nos. 6,809,078; 6,713,063; 6,686,452; 6,657,046; 6,645,497; 6,642,030; 6,639,054; 6,620,988; 6,593,293; 6,555,655; 6,538,109; 6,537,756; 6,521,442; 6,501,009; 6,468,523; 6,326,351; 6,313,378; 6,284,949; 6,281,016; 6,248,536; 6,242,241; 6,221,649; 6,177,615; 6,156,573; 6,153,814; 6,110,464; 6,093,695; 6,063,756; 6,063,597; 6,023,013; 5,959,091; 5,942,664; 5,942,658; 5,880,275; 5,763,245 and 5,763,241.

D. Male Sterility

Genetic male sterility is available in soybeans and, although not required for crossing soybean plants, can increase the efficiency with which hybrids are made, as it eliminates the need to physically emasculate the soybean plant used as a female in a given cross. (Brim and Stuber, *Crop Sci.*, 13:528-530, 1973). Herbicide-inducible male sterility systems have also been described in, for example, U.S. Pat. No. 6,762,344.

When one desires to employ male-sterility systems, it may be beneficial to also utilize one or more male-fertility restorer genes. For example, when cytoplasmic male sterility (CMS) is used, hybrid seed production requires three inbred lines: (1) a cytoplasmically male-sterile line having a CMS cytoplasm; (2) a fertile inbred with normal cytoplasm, which is isogenic with the CMS line for nuclear genes ("maintainer line"); and (3) a distinct, fertile inbred with normal cytoplasm, carrying a fertility restoring gene ("restorer" line). The CMS line is propagated by pollination with the maintainer line, and all of the progeny are male sterile, as the CMS cytoplasm is derived from the female parent. These male sterile plants can then be efficiently employed as the female parent in hybrid crosses with the restorer line,

without the need for physical emasculation of the male reproductive parts of the female parent.

The presence of a male-fertility restorer gene results in the production of fully fertile F_1 hybrid progeny. If no restorer gene is present in the male parent, male-sterile hybrids are obtained. Such hybrids are useful where the vegetative tissue of the soybean plant is utilized, but in many cases the seeds will be deemed the most valuable portion of the crop, so fertility of the hybrids in these crops must be restored. Therefore, one aspect of the current invention concerns plants of the soybean variety 01098164 comprising a genetic locus capable of restoring male fertility in an otherwise male-sterile plant. Examples of male-sterility genes and corresponding restorers which could be employed with the plants of the invention are well known to those of skill in the art of plant breeding, see, for example, U.S. Pat. Nos. 5,530,191 and 5,684,242, the disclosures of which are each specifically incorporated herein by reference in their entirety.

E. Modified Fatty Acid, Phytate, and Carbohydrate Metabolism

Genes may be used conferring modified fatty acid metabolism. For example, stearyl-ACP desaturase genes may be used, see Knutzon et al. (*Proc. Natl. Acad. Sci. USA*, 89:2624-2628, 1992). Various fatty acid desaturases have also been described. McDonough et al. describe a *Saccharomyces cerevisiae* OLE1 gene encoding $\Delta 9$ -fatty acid desaturase, an enzyme which forms the monounsaturated palmitoleic (16:1) and oleic (18:1) fatty acids from palmitoyl (16:0) or stearyl (18:0) CoA (*J. Biol. Chem.*, 267(9): 5931-5936, 1992). Fox et al. describe a gene encoding a stearyl-acyl carrier protein delta-9 desaturase from castor (*Proc. Natl. Acad. Sci. USA*, 90(6):2486-2490, 1993). Reddy et al. describe $\Delta 6$ - and $\Delta 12$ -desaturases from the cyanobacteria *Synechocystis* responsible for the conversion of linoleic acid (18:2) to gamma-linolenic acid (18:3 gamma) (*Plant Mol. Biol.*, 22(2):293-300, 1993). Arondel et al. describe a gene from *Arabidopsis thaliana* that encodes an omega-3 desaturase has been identified (*Science*, 258(5086):1353-1355, 1992). Plant $\Delta 9$ -desaturases as well as soybean and *Brassica* $\Delta 15$ -desaturases have also been described, see PCT Application Publication No. WO 91/13972 and European Patent Application Publication No. EP0616644, respectively. U.S. Pat. No. 7,622,632 describes fungal $\Delta 15$ -desaturases and their use in plants. European Patent Application Publication No. EP1656449 describes $\Delta 6$ -desaturases from *Primula* as well as soybean plants having increased stearidonic acid (SDA, 18:4) content. U.S. Pat. No. 8,378,186 describes expression of transgenic desaturase enzymes in corn plants, and improved fatty acid profiles resulting therefrom.

Modified oil production is disclosed in, for example, U.S. Pat. Nos. 6,444,876; 6,426,447; and 6,380,462. High oil production is disclosed in, for example, U.S. Pat. Nos. 6,495,739; 5,608,149; 6,483,008; and 6,476,295. Modified fatty acid content is disclosed in, for example, U.S. Pat. Nos. 6,828,475; 6,822,141; 6,770,465; 6,706,950; 6,660,849; 6,596,538; 6,589,767; 6,537,750; 6,489,461 and 6,459,018.

Phytate metabolism may also be modified by introduction of a phytase-encoding gene to enhance breakdown of phytate, adding more free phosphate to the transformed plant. For example, see Van Hartingsveldt et al. (*Gene*, 127:87-94, 1993), for a disclosure of the nucleotide sequence of an *Aspergillus niger* phytase gene. For example, this could be accomplished in soybean plants by cloning and then reintroducing DNA associated with the single allele that

is responsible for soybean mutants characterized by low levels of phytic acid. See Raboy et al. (*Plant Physiol.*, 124(1):355-368, 2000).

A number of genes are known that may be used to alter carbohydrate metabolism. For example, plants may be transformed with a gene coding for an enzyme that alters the branching pattern of starch. For example, Shiroza et al. (*J. Bacteriol.*, 170:810-816, 1988) describe a nucleotide sequence of the *Streptococcus mutans* fructosyltransferase gene; Steinmetz et al. (*Mol. Gen. Genet.*, 20:220-228, 1985) describe a nucleotide sequence of the *Bacillus subtilis* levansucrase gene; Pen et al. (*Biotechnology*, 10:292-296, 1992) describe production of transgenic plants that express *Bacillus licheniformis* α -amylase; Elliot et al. (*Plant Molec. Biol.*, 21:515-524, 1993) describe nucleotide sequences of tomato invertase genes; Sergaard et al. (*J. Biol. Chem.*, 268:22480, 1993) describe site-directed mutagenesis of a barley α -amylase gene; and Fisher et al. (*Plant Physiol.*, 102:1045-1046, 1993) describe maize endosperm starch branching enzyme II. The Z10 gene encoding a 10 kD zein storage protein from maize may also be used to alter the quantities of 10 kD zein in the cells relative to other components (Kirihaara et al., *Gene*, 71(2):359-370, 1988).

F. Resistance to Abiotic Stress

Abiotic stress includes dehydration or other osmotic stress, salinity, high or low light intensity, high or low temperatures, submergence, exposure to heavy metals, and oxidative stress. Delta-pyrroline-5-carboxylate synthetase (P5CS) from mothbean has been used to provide protection against general osmotic stress. Mannitol-1-phosphate dehydrogenase (mt1D) from *E. coli* has been used to provide protection against drought and salinity. Choline oxidase (codA from *Arthrobacter globiformis*) can protect against cold and salt. *E. coli* choline dehydrogenase (betA) provides protection against salt. Additional protection from cold can be provided by omega-3-fatty acid desaturase (fad7) from *Arabidopsis thaliana*. Trehalose-6-phosphate synthase and levan sucrase (SacB) from yeast and *Bacillus subtilis*, respectively, can provide protection against drought (summarized from Annex II Genetic Engineering for Abiotic Stress Tolerance in Plants, Consultative Group On International Agricultural Research Technical Advisory Committee). Overexpression of superoxide dismutase can be used to protect against superoxides, see U.S. Pat. No. 5,538,878.

G. Additional Traits

Additional traits can be introduced into the soybean variety of the present invention. A non-limiting example of such a trait is a coding sequence which decreases RNA and/or protein levels. The decreased RNA and/or protein levels may be achieved through RNAi methods, such as those described in U.S. Pat. No. 6,506,559.

Another trait that may find use with the soybean variety of the invention is a sequence which allows for site-specific recombination. Examples of such sequences include the FRT sequence used with the FLP recombinase (Zhu and Sadowski, *J. Biol. Chem.*, 270:23044-23054, 1995) and the LOX sequence used with CRE recombinase (Sauer, *Mol. Cell. Biol.*, 7:2087-2096, 1987). The recombinase genes can be encoded at any location within the genome of the soybean plant and are active in the hemizygous state.

In certain embodiments soybean plants may be made more tolerant to or more easily transformed with *Agrobacterium tumefaciens*. For example, expression of p53 and iap, two baculovirus cell-death suppressor genes, inhibited tissue necrosis and DNA cleavage. Additional targets may include plant-encoded proteins that interact with the *Agrobacterium* Vir genes; enzymes involved in plant cell wall formation;

and histones, histone acetyltransferases and histone deacetylases (reviewed in Gelvin, *Microbiology & Mol. Biol. Reviews*, 67:16-37, 2003).

In addition to the modification of oil, fatty acid, or phytate content described above, certain embodiments may modify the amounts or levels of other compounds. For example, the amount or composition of antioxidants can be altered. See, for example, U.S. Pat. Nos. 6,787,618 and 7,154,029 and International Patent Application Publication No. WO 00/68393, which disclose the manipulation of antioxidant levels, and International Patent Application Publication No. WO 03/082899, which discloses the manipulation of an antioxidant biosynthetic pathway.

Additionally, seed amino acid content may be manipulated. U.S. Pat. No. 5,850,016 and International Patent Application Publication No. WO 99/40209 disclose the alteration of the amino acid compositions of seeds. U.S. Pat. Nos. 6,080,913 and 6,127,600 disclose methods of increasing accumulation of essential amino acids in seeds.

U.S. Pat. No. 5,559,223 describes synthetic storage proteins of which the levels of essential amino acids can be manipulated. International Patent Application Publication No. WO 99/29882 discloses methods for altering amino acid content of proteins. International Patent Application Publication No. WO 98/20133 describes proteins with enhanced levels of essential amino acids. International Patent Application Publication No. WO 98/56935 and U.S. Pat. Nos. 6,346,403; 6,441,274; and 6,664,445 disclose plant amino acid biosynthetic enzymes. International Patent Application Publication No. WO 98/45458 describes synthetic seed proteins having a higher percentage of essential amino acids than wild-type.

U.S. Pat. No. 5,633,436 discloses plants comprising a higher content of sulfur-containing amino acids; U.S. Pat. No. 5,885,801 discloses plants comprising a high threonine content; U.S. Pat. Nos. 5,885,802 and 5,912,414 disclose plants comprising a high methionine content; U.S. Pat. No. 5,990,389 discloses plants comprising a high lysine content; U.S. Pat. No. 6,459,019 discloses plants comprising an increased lysine and threonine content; International Patent Application Publication No. WO 98/42831 discloses plants comprising a high lysine content; International Patent Application Publication No. WO 96/01905 discloses plants comprising a high threonine content; and International Patent Application Publication No. WO 95/15392 discloses plants comprising a high lysine content.

Origin and Breeding History of an Exemplary Single Locus Converted Plant

It is known to those of skill in the art that, by way of the technique of backcrossing, one or more traits may be introduced into a given variety while otherwise retaining essentially all of the traits of that variety. An example of such backcrossing to introduce a trait into a starting variety is described in U.S. Pat. No. 6,140,556, the entire disclosure of which is specifically incorporated herein by reference. The procedure described in U.S. Pat. No. 6,140,556 can be summarized as follows: The soybean variety known as Williams '82 [*Glycine max* L. Merr.] (Reg. No. 222, PI 518671) was developed using backcrossing techniques to transfer a locus comprising the Rps₁ gene to the variety Williams (Bernard and Cremeens, *Crop Sci.*, 28:1027-1028, 1988). Williams '82 is a composite of four resistant lines from the BC₆F₃ generation, which were selected from 12 field-tested resistant lines from Williams x Kingwa. The variety Williams was used as the recurrent parent in the

backcross and the variety Kingwa was used as the source of the Rps₁ locus. This gene locus confers resistance to 19 of the 24 races of the fungal agent *Phytophthora* root rot.

The F₁ or F₂ seedlings from each backcross round were tested for resistance to the fungus by hypocotyl inoculation using the inoculum of race 5. The final generation was tested using inoculum of races 1 to 9. In a backcross such as this, in which the desired characteristic being transferred to the recurrent parent is controlled by a major gene which can be readily evaluated during the backcrossing, it is common to conduct enough backcrosses to avoid testing individual progeny for specific traits such as yield in extensive replicated tests. In general, four or more backcrosses are used when there is no evaluation of the progeny for specific traits. As in this example, lines with the phenotype of the recurrent parent may be composited without the usual replicated tests for traits, such as yield, protein, or oil percentage, in the individual lines.

The variety Williams '82 is comparable to the recurrent parent variety Williams in its traits except resistance to *Phytophthora* rot. For example, both varieties have a relative maturity of 38, indeterminate stems, white flowers, brown pubescence, tan pods at maturity, and shiny yellow seeds with black to light black hila.

Tissue Cultures and In Vitro Regeneration of Soybean Plants

A further aspect of the invention relates to tissue cultures of the soybean variety designated 01098164. As used herein, the term "tissue culture" indicates a composition comprising isolated cells of the same or a different type or a collection of such cells organized into parts of a plant. Exemplary types of tissue cultures are protoplasts, calli, and plant cells that are intact in plants or parts of plants, such as embryos, pollen, flowers, leaves, roots, root tips, anthers, and the like. In one embodiment, the tissue culture comprises embryos, protoplasts, meristematic cells, pollen, leaves, or anthers.

Exemplary procedures for preparing tissue cultures of regenerable soybean cells and regenerating soybean plants therefrom are disclosed in U.S. Pat. Nos. 4,992,375; 5,015,580; 5,024,944; and 5,416,011, each of which are specifically incorporated herein by reference in their entirety.

An important ability of a tissue culture is the capability to regenerate fertile plants. This allows, for example, transformation of the tissue culture cells followed by regeneration of transgenic plants. For transformation to be efficient and successful, DNA must be introduced into cells that give rise to plants or germ-line tissue.

Soybeans typically are regenerated via two distinct processes: shoot morphogenesis and somatic embryogenesis (Finer, Cheng, Verma, "Soybean transformation: Technologies and progress," In: *Soybean: Genetics, Molecular Biology and Biotechnology*, CAB Intl, Verma and Shoemaker (ed), Wallingford, Oxon, UK, 250-251, 1996). Shoot morphogenesis is the process of shoot meristem organization and development. Shoots grow out from a source tissue and are excised and rooted to obtain an intact plant. During somatic embryogenesis, an embryo (similar to the zygotic embryo), containing both shoot and root axes, is formed from somatic plant tissue. An intact plant rather than a rooted shoot results from the germination of the somatic embryo.

Shoot morphogenesis and somatic embryogenesis are different processes and the specific route of regeneration is primarily dependent on the explant source and media used for tissue culture manipulations. While the systems are

different, both systems show variety-specific responses in which some lines are more responsive to tissue culture manipulations than others. A line that is highly responsive in shoot morphogenesis may not generate many somatic embryos. Lines that produce large numbers of embryos during an 'induction' step may not give rise to rapidly-growing proliferative cultures. Therefore, it may be desired to optimize tissue culture conditions for each soybean line. These optimizations may readily be carried out by one of skill in the art of tissue culture through small-scale culture studies. In addition to line-specific responses, proliferative cultures can be observed with both shoot morphogenesis and somatic embryogenesis. Proliferation is beneficial for both systems as it allows a single, transformed cell to multiply to the point that it will contribute to germ-line tissue.

Shoot morphogenesis was first reported by Wright et al. (*Plant Cell Reports*, 5:150-154, 1986) as a system from which shoots were obtained de novo from cotyledonary nodes of soybean seedlings. The shoot meristems were formed subepidermally and morphogenic tissue could proliferate on a medium containing benzyl adenine (BA). This system can be used for transformation if the subepidermal, multicellular origin of the shoots is recognized and proliferative cultures are utilized. The idea is to target tissue that will give rise to new shoots and proliferate those cells within the meristematic tissue to lessen problems associated with chimerism. Formation of chimeras, as a result of transforming only a single cell in a meristem, is problematic if the transformed cell is not adequately proliferated and does not give rise to germ-line tissue. Once the system is well understood and reproduced satisfactorily, it can be used as one target tissue for soybean transformation.

Somatic embryogenesis in soybean was first reported by Christianson et al. (*Science*, 222:632-634, 1983) as a system in which embryogenic tissue was initially obtained from the zygotic embryo axis. These embryogenic cultures were proliferative but the repeatability of the system was low and the origin of the embryos was not reported. Later histological studies of a different proliferative embryogenic soybean culture showed that proliferative embryos were of apical or surface origin with a small number of cells contributing to embryo formation. The origin of primary embryos, the first embryos derived from the initial explant, is dependent on the explant tissue and the auxin levels in the induction medium (Hartweck et al., *In Vitro Cell. Develop. Bio.*, 24:821-828, 1988). With proliferative embryonic cultures, single cells or small groups of surface cells of the 'older' somatic embryos form the 'newer' embryos.

Embryogenic cultures can also be used successfully for regeneration, including regeneration of transgenic plants, if the origin of the embryos is recognized and the biological limitations of proliferative embryogenic cultures are understood. Biological limitations include the difficulty in developing proliferative embryogenic cultures and reduced fertility problems (culture-induced variation) associated with plants regenerated from long-term proliferative embryogenic cultures. Some of these problems are accentuated in prolonged cultures. The use of more recently cultured cells may decrease or eliminate such problems.

Definitions

In the description and table, a number of terms are used. In order to provide a clear and consistent understanding of the specification and claims, the following definitions are provided:

A: When used in conjunction with the word "comprising" or other open language in the claims, the words "a" and "an" denote "one or more."

About: Refers to embodiments or values that include the standard deviation of the mean for a given item being measured.

Allele: Any of one or more alternative forms of a locus. In a diploid cell or organism, the two alleles of a given locus occupy corresponding loci on a pair of homologous chromosomes.

Aphids: Aphid resistance in greenhouse screening is scored based on foliar symptoms and number of aphids using a 1 to 9 scale. "Resistant" (R) corresponds to a rating between "1" and "3.9," inclusive. "Moderately Resistant" (MR) corresponds to a rating between "4.0" and "5.9," inclusive. "Moderately Susceptible to Moderately Resistant" (MS-MR) corresponds to a rating between "6.0" and "6.9," inclusive. "Susceptible" (S) corresponds to a rating between "7.0" and "9.0," inclusive.

Asian Soybean Rust (ASR): ASR may be visually scored based on a 1 to 5 scale. A score of "1" indicates "immune." A score of "2" indicates that the leaves exhibit red/brown lesions over less than 50% of surface. A score of "3" indicates that the leaves exhibit red/brown lesions over greater than 50% of surface. A score of "4" indicates that the leaves exhibit tan lesions over less than 50% of surface. A score of "5" indicates that the leaves exhibit tan lesions over greater than 50% of surface. Resistance to ASR may be characterized phenotypically as well as genetically. Soybean plants phenotypically characterized as resistant to ASR typically exhibit red/brown lesions covering less than 25% of the leaf. Genetic characterization of ASR resistance may be carried out, for example, by identifying the presence in a soybean plant of one or more genetic markers linked to the ASR resistance.

Backcrossing: A process in which a breeder repeatedly crosses hybrid progeny, for example a first generation hybrid (F₁), back to one of the parents of the hybrid progeny. Backcrossing can be used to introduce one or more single locus conversions from one genetic background into another.

Brown Stem Rot (BSR): The greenhouse score is based on the incidence and severity of both foliar symptoms and pith discoloration and scores are converted to a 1 to 9 scale. "Resistant" (R) corresponds to a rating less than "3.5." "Moderately Resistant" (MR) corresponds to a rating between "3.5" and "4.4," inclusive. "Moderately Resistant-Moderately Susceptible" (MR/MS) corresponds to a rating between "4.5" and "5.4," inclusive. "Moderately Susceptible" (MS) corresponds to a rating between "5.5" and "6.4," inclusive. "Susceptible" (S) corresponds to a rating of "6.5" or greater.

Chloride Sensitivity: The greenhouse score is based on the average severity of tissue chlorosis/necrosis and is converted to a 1-9 scale. "Tolerant" (T) corresponds to a rating less than "2.6." "Moderately Tolerant" (MT) corresponds to a rating between "2.6" and "4.0," inclusive. "Sensitive" (S) corresponds to a rating greater than "4.0."

Crossing: The mating of two parent plants.

Cross-pollination: Fertilization by the union of two gametes from different plants.

Emasculate: The removal of plant male sex organs or the inactivation of the organs with a cytoplasmic or nuclear genetic factor or a chemical agent conferring male sterility.

Enzymes: Molecules which can act as catalysts in biological reactions.

F₁ Hybrid: The first generation progeny of the cross of two nonisogenic plants.

Fatty Acids: Are measured and reported as a percent of the total oil content. In addition to the typical composition of fatty acids in commodity soybeans, some soybean varieties have modified profiles. Low linolenic acid soybean oil as defined herein contains 3% or less linolenic acid. Mid oleic acid soybean oil as defined herein typically contains 50-60% oleic acid. High oleic soybean oil as defined herein typically contains 75% or greater oleic acid. Stearidonic acid levels are typically 0% in commodity soybeans.

Frog Eye Leaf Spot (FELS): Greenhouse assay reaction scores are based on foliar symptom severity and are measured using a 1-9 scale. "Resistant" (R) corresponds to a rating less than "3.5." "Moderately Resistant" (MR) corresponds to a rating between "3.5" and "4.4," inclusive. "Moderately Resistant-Moderately Susceptible" (MR/MS) corresponds to a rating between "4.5" and "5.4," inclusive. "Moderately Susceptible" (MS) corresponds to a rating between "5.5" and "6.4," inclusive. "Susceptible" (S) corresponds to a rating of "6.5" or greater.

Genotype: The genetic constitution of a cell or organism.

Haploid: A cell or organism having one set of the two sets of chromosomes in a diploid.

Iron-Deficiency Chlorosis (IDE=early; IDL=late): Iron-deficiency chlorosis is scored based on visual observations using 1 to 9 scale. A score of "1" indicates that no stunting of the plants or yellowing of the leaves was observed. A score of "9" indicates that the plants are dead or dying as a result of iron-deficiency chlorosis. A score of "5" means that plants display intermediate health and some observable leaf yellowing.

Linkage: A phenomenon wherein alleles on the same chromosome tend to segregate together more often than expected by chance if their transmission was independent.

Linolenic Acid Content (LLN): Low-linolenic acid soybean oil contains 3% or less linolenic acid. Traditional soybean oil contains approximately 8% linolenic acid.

Marker: A readily detectable phenotype, preferably inherited in co-dominant fashion (both alleles at a locus in a diploid heterozygote are readily detectable), with no environmental variance component, i.e., heritability of 1.

Maturity Date (MAT): Plants are considered mature when 95% of the pods have reached their mature color. The maturity date is typically described in measured days after August 31 in the northern hemisphere.

Moisture (MST): The average percentage moisture in the seeds of a variety.

Oil or Oil Percent: Seed oil content is measured and reported on a percentage basis.

Or: As used herein is meant to mean "and/or" and be interchangeable therewith unless explicitly indicated to refer to the alternative only.

Phenotype: The detectable characteristics of a cell or organism, which are the manifestation of gene expression.

Phenotypic Score (PSC): The phenotypic score is a visual rating of the general appearance of the variety. All visual traits are considered in the score, including healthiness, standability, appearance, and freedom from disease. Ratings are scored as 1 being poor to 9 being excellent.

Phytophthora Allele: Susceptibility or resistance to *Phytophthora* root rot races is affected by alleles such as Rps1a (denotes resistance to Races 1, 2, 10, 11, 13-18, 24, 26, 27, 31, 32, and 36); Rps1c (denotes resistance to Races 1-3, 6-11, 13, 15, 17, 21, 23, 24, 26, 28-30, 32, 34 and 36); Rps1k (denotes resistance to Races 1-11, 13-15, 17, 18, 21-24, 26, 36 and 37); Rps2 (denotes resistance to Races 1-5, 9-29, 33,

34 and 36-39); Rps3a (denotes resistance to Races 1-5, 8, 9, 11, 13, 14, 16, 18, 23, 25, 28, 29, 31-35); Rps6 (denotes resistance to Races 1-4, 10, 12, 14-16, 18-21 and 25); and Rps7 (denotes resistance to Races 2, 12, 16, 18, 19, 33, 35 and 36).

Phytophthora Root Rot (PRR): Disorder of which the most recognizable symptom is stem rot. Brown discoloration ranges below the soil line and up to several inches above the soil line. The leaves often turn yellow, dull green and/or gray and may become brown and wilted, but remain attached to the plant.

Phytophthora Tolerance: Tolerance to *Phytophthora* root rot is rated in the greenhouse assay based on a 1 to 9 scale. "Resistant" (R) corresponds to a rating less than "3.5." "Moderately Resistant" (MR) corresponds to a rating between "3.5" and "4.4," inclusive. "Moderately Resistant-Moderately Susceptible" (MR/MS) corresponds to a rating between "4.5" and "5.4," inclusive. "Moderately Susceptible" (MS) corresponds to a rating between "5.5" and "6.4," inclusive. "Susceptible" (S) corresponds to a rating of "6.5" or greater. Note that a score between "1" and "2" may indicate resistance to *Phytophthora* and therefore not be a true reflection of high tolerance to *Phytophthora*.

Plant Height (PHT): Plant height is taken from the top of soil to the top node of the plant and is measured in inches.

Predicted Relative Maturity (PRM): The maturity grouping designated by the soybean industry over a given growing area. This FIGURE is generally divided into tenths of a relative maturity group. Within narrow comparisons, the difference of a tenth of a relative maturity group equates very roughly to a day difference in maturity at harvest.

Protein (PRO), or Protein Percent: Seed protein content is measured and reported on a percentage basis.

Regeneration: The development of a plant from tissue culture.

Relative Maturity: The maturity grouping designated by the soybean industry over a given growing area. This FIGURE is generally divided into tenths of a relative maturity group. Within narrow comparisons, the difference of a tenth of a relative maturity group equates very roughly to a day difference in maturity at harvest.

Seed Protein Peroxidase Activity: Seed protein peroxidase activity is defined as a chemical taxonomic technique to separate varieties based on the presence or absence of the peroxidase enzyme in the seed coat. There are two types of soybean varieties, those having high peroxidase activity (dark red color) and those having low peroxidase activity (no color).

Seed Weight (SWT): Soybean seeds vary in size; therefore, the number of seeds required to make up one pound also varies. This affects the pounds of seed required to plant a given area and can also impact end uses. "SW100" equals the weight in grams of 100 seeds.

Seed Yield (Bushels/Acre): The yield in bushels/acre is the actual yield of the grain at harvest.

Seedling Vigor Rating (SDV): General health of the seedling that is measured on a 1 to 9 scale in which "1" is "best" and "9" is "worst."

Seeds per Pound: Soybean seeds vary in size; therefore, the number of seeds required to make up one pound also varies. This affects the pounds of seed required to plant a given area and can also impact end uses.

Self-pollination: The transfer of pollen from the anther to the stigma of the same plant.

Shattering: The amount of pod dehiscence prior to harvest. Pod dehiscence involves seeds falling from the pods to the soil. This is a visual score from 1 to 9 comparing all

genotypes within a given test. A score of “1” indicates that the pods have not opened and no seeds have fallen out. A score of “5” indicates that approximately 50% of the pods have opened, with seeds falling to the ground. A score of “9” indicates that 100% of the pods are opened.

Single Locus Converted (Conversion) Plant: Plants that are developed by a plant breeding technique called backcrossing or by genome editing of a locus, in which essentially all of the morphological and physiological characteristics of a soybean variety are recovered in addition to the characteristics of the locus transferred into the variety via the backcrossing technique or by genetic transformation. It is understood that once introduced into any soybean plant genome, a locus that is transgenic in origin (transgene), can be introduced by backcrossing as with any other locus.

Southern Root Knot Nematode (SRKN): Greenhouse assay reaction scores are based on a gall index, ultimately converted to a 1 to 9 scale. “Resistant” (R) corresponds to a rating less than “6.12.” “Moderately Resistant” (MR) corresponds to a rating between “6.12” and “6.6,” inclusive. “Moderately Resistant to Moderately Susceptible” (MR-MS) corresponds to a rating between “6.6” and “7.4,” inclusive. “Susceptible” (S) corresponds to a rating greater than “7.4.”

Southern Stem Canker (STC): Greenhouse assay scoring is based on percentage of dead plants (DP). This percentage is converted to a 1 to 9 scale: “1” corresponds to no DP. “2” corresponds to less than 10% DP. “3” corresponds to between 10% and 30%, inclusive, DP. “4” corresponds to between 31% and 40%, inclusive, DP. “5” corresponds to between 41% and 50%, inclusive, DP. “6” corresponds to between 51% and 60%, inclusive, DP. “7” corresponds to between 61%-70%, inclusive, DP. “8” corresponds to between 71% and 90%, inclusive, DP. “9” corresponds to between 91% and 100%, inclusive, DP. “Resistant” (R) corresponds to a rating less than “3.5.” “Moderately Resistant” (MR) corresponds to a rating between “3.5” and “4.4,” inclusive. “Moderately Resistant-Moderately Susceptible” (MR/MS) corresponds to a rating between “4.5” and “5.4,” inclusive. “Moderately Susceptible” (MS) corresponds to a rating between “5.5” and “6.4,” inclusive. “Susceptible” (S) corresponds to a rating of “6.5” or greater.

Soybean Cyst Nematode (SCN): Greenhouse screening scores are based on a female index % of Lee 74. “Resistant” (R) corresponds to a rating less than 10%. “Moderately Resistant” (MR) corresponds to a rating between 10% and 21.9%, inclusive. “Moderately Resistant to Moderately Susceptible” (MR-MS) corresponds to a rating between 22% and 39.9%, inclusive. “Susceptible” (S) corresponds to a rating greater than 39.9%.

Stearate: A fatty acid in soybean seeds measured and reported as a percent of the total oil content.

Substantially Equivalent: A characteristic that, when compared, does not show a statistically significant difference from the mean, e.g., $p=0.05$.

Sudden Death Syndrome: Leaf symptoms first appear as bright yellow chlorotic spots with progressive development of brown necrotic areas and eventual leaflet drop. Greenhouse screening plants are scored based on foliar symptom severity using a 1 to 9 scale. “Resistant” (R) corresponds to a rating less than “3.” “Moderately Resistant” (MR) corresponds to a rating between “3.0” and “4.9,” inclusive. “Moderately Susceptible” (MS) corresponds to a rating between “5.0” and “6.9,” inclusive. “Susceptible” (S) corresponds to a rating between “7.0” and “8.0,” inclusive. “Highly Susceptible” (HS) corresponds to a rating greater than “8.”

Tissue Culture: A composition comprising isolated cells of the same or a different type or a collection of such cells organized into parts of a plant.

Transgene: A genetic locus comprising a sequence which has been introduced into the genome of a soybean plant by transformation or site-specific recombination.

DEPOSIT INFORMATION

A deposit of the soybean variety 01098164, which is disclosed herein above and referenced in the claims, has been made with the Provasoli-Guillard National Center for Marine Algae and Microbiota (NCMA) at Bigelow Laboratory for Ocean Sciences, 60 Bigelow Drive, East Boothbay, ME 04544 USA. The date of deposit is May 4, 2022, and the accession number for those deposited seeds of soybean variety 01098164 is NCMA Accession No. 202205056. All restrictions upon availability of the deposit to the public will be irrevocably removed upon granting of the patent, and the deposit is intended to meet all of the requirements of the Budapest Treaty and 37 C.F.R. §§ 1.801-1.809. Access to the deposit will be available during the pendency of the application to the Commissioner of Patents and Trademarks and persons determined by the Commissioner to be entitled thereto upon request. The deposit has been accepted under the Budapest Treaty and will be maintained in the NCMA Depository, which is a public depository, for a period of 30 years, or 5 years after the most recent request, or for the enforceable life of the patent, whichever is longer, and will be replaced if it becomes nonviable during that period. Applicant does not waive any infringement of their rights granted under this patent or under the Plant Variety Protection Act (7 U.S.C. § 2321 et seq.).

All of the compositions and methods disclosed and claimed herein can be made and executed without undue experimentation in light of the present disclosure. While the compositions and methods of this invention have been described in terms of the foregoing illustrative embodiments, it will be apparent to those of skill in the art that variations, changes, modifications, and alterations may be applied to the composition, methods, and in the steps or in the sequence of steps of the methods described herein, without departing from the true concept, spirit, and scope of the invention. More specifically, it will be apparent that certain agents that are both chemically and physiologically related may be substituted for the agents described herein while the same or similar results would be achieved. All such similar substitutes and modifications apparent to those skilled in the art are deemed to be within the spirit, scope, and concept of the invention as defined by the appended claims.

The references cited herein, to the extent that they provide exemplary procedural or other details supplementary to those set forth herein, are specifically incorporated herein by reference.

What is claimed:

1. A plant of soybean variety 01098164, wherein representative seed of said soybean variety have been deposited under NCMA Accession No. 202205056.
2. A plant part of the plant of claim 1, wherein the plant part comprises at least one cell of said plant.
3. A seed of soybean variety 01098164, wherein representative seed of said soybean variety have been deposited under NCMA Accession No. 202205056.

33

4. A method of producing a soybean seed, the method comprising crossing the plant of claim 1 with itself or a second soybean plant to produce said soybean seed.

5. The method of claim 4, the method further comprising crossing the plant of soybean variety 01098164 with a second, nonisogenic soybean plant to produce said soybean seed.

6. An F_1 soybean seed produced by the method of claim 5.

7. An F_1 soybean plant produced by growing the F_1 soybean seed of claim 6.

8. A composition comprising the seed of claim 3 comprised in plant seed growth media.

9. The composition of claim 8, wherein the plant seed growth media is soil or a synthetic cultivation medium.

10. A plant of soybean variety 01098164 further comprising a single locus conversion, wherein said plant otherwise comprises all of the morphological and physiological characteristics of said soybean variety, and wherein representative seed of said soybean variety have been deposited under NCMA Accession No. 202205056.

11. The plant of claim 10, wherein the single locus conversion comprises a transgene.

12. A seed that produces the plant of claim 10.

13. The seed of claim 12, wherein the single locus conversion comprises a nucleic acid sequence that enables site-specific genetic recombination or confers a trait selected from the group consisting of male sterility, herbicide tolerance, insect resistance, pest resistance, disease resistance, modified fatty acid metabolism, abiotic stress resistance, altered seed amino acid composition, and modified carbohydrate metabolism.

14. The seed of claim 13, wherein the single locus conversion that confers herbicide tolerance confers tolerance to benzonitrile herbicides, cyclohexanedione herbicides, imidazolinone herbicides, phenoxy herbicides, sulfonylurea herbicides, triazine herbicides, 1-aminocyclopropane-1-carboxylic acid synthase-inhibiting

34

herbicides, 4-hydroxyphenylpyruvate dioxygenase-inhibiting herbicides, acetolactate synthase-inhibiting herbicides, protoporphyrinogen oxidase-inhibiting herbicides, 2,4-dichlorophenoxyacetic acid (2,4-D), bromoxynil, dicamba, glufosinate, glyphosate, nicosulfuron, or quizalofop-p-ethyl.

15. The seed of claim 12, wherein the single locus conversion comprises a transgene.

16. The method of claim 5, the method further comprising:

(a) crossing a plant grown from said soybean seed with itself or a different soybean plant to produce seed of a progeny plant of a subsequent generation;

(b) growing a progeny plant of a subsequent generation from said seed of the progeny plant of the subsequent generation and crossing the progeny plant of the subsequent generation with itself or a second plant to produce seed of a progeny plant of a further subsequent generation; and

(c) repeating step (b) with sufficient inbreeding to produce seed of an inbred soybean plant that is derived from soybean variety 01098164.

17. The method of claim 16, the method further comprising crossing a plant grown from said seed of the inbred soybean plant that is derived from soybean variety 01098164 with a nonisogenic plant to produce seed of a hybrid soybean plant that is derived from soybean variety 01098164.

18. A method of producing a commodity plant product, the method comprising producing the commodity plant product from the plant of claim 1.

19. The method of claim 18, wherein the commodity plant product is selected from the group consisting of protein concentrate, protein isolate, grain, soybean hulls, meal, flour, and oil.

20. A commodity plant product that is produced by the method of claim 18, wherein the commodity plant product comprises at least one cell of soybean variety 01098164.

* * * * *